



# Fundamentals of Optics

## Lecturer

**Yanjun LIU (刘言军)**

E-mail: [yjliu@sustech.edu.cn](mailto:yjliu@sustech.edu.cn)

Room 239, South Tower, College of Engineering  
Department of Electrical and Electronic Engineering  
Southern University of Science and Technology  
Shenzhen, Guangdong, China

## Teaching Assistant

**Zhenming WANG (王振铭)**

E-mail: [12150025@mail.sustech.edu.cn](mailto:12150025@mail.sustech.edu.cn)

Room 338, South Tower, College of Engineering

**Qing-an TU (涂清安)**

E-mail: [12231044@mail.sustech.edu.cn](mailto:12231044@mail.sustech.edu.cn)

Room 320, South Tower, College of Engineering

**Q&A time: 2-4 pm on Monday**

Room 239, South Tower, College of Engineering

## Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

## Reference books

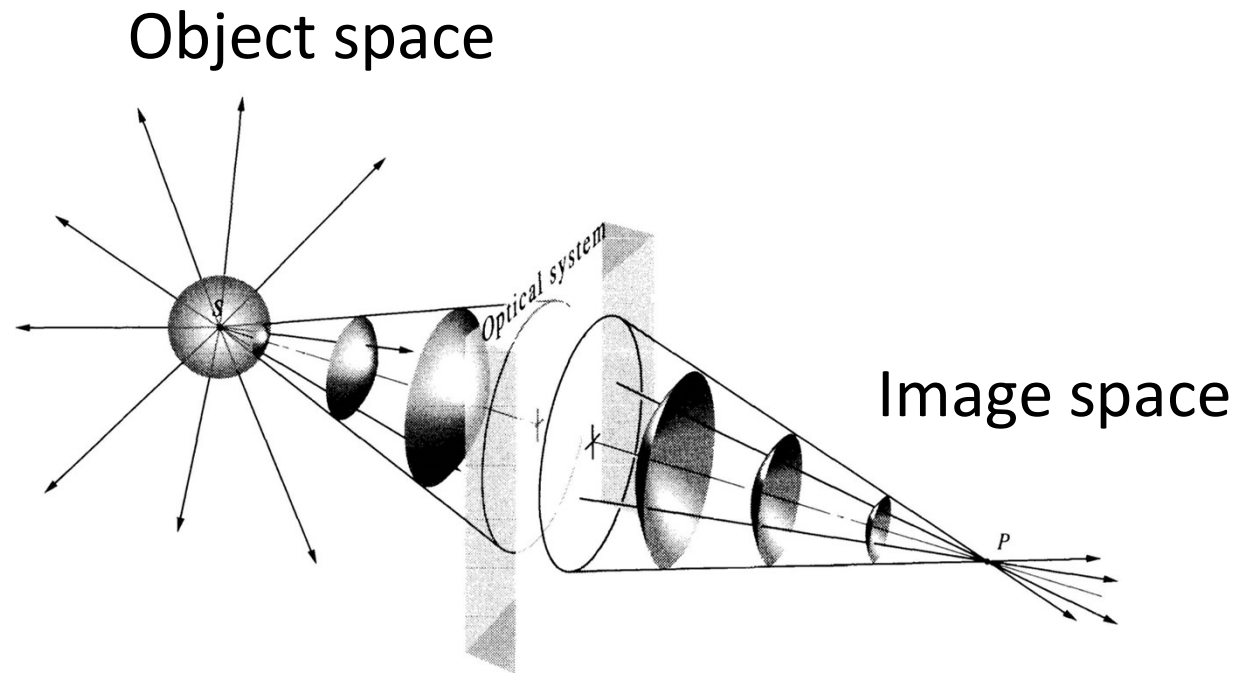
Optics, by Eugene Hecht  
光学, 赵凯华、钟锡华

# **CHAPTER 4**

## **GEOMETRICAL OPTICS**

# 4.1 Introductory Remarks

A point from which a portion of a spherical wave diverges, or one toward which the wave segment converges, is known as a focus of the bundle of rays.

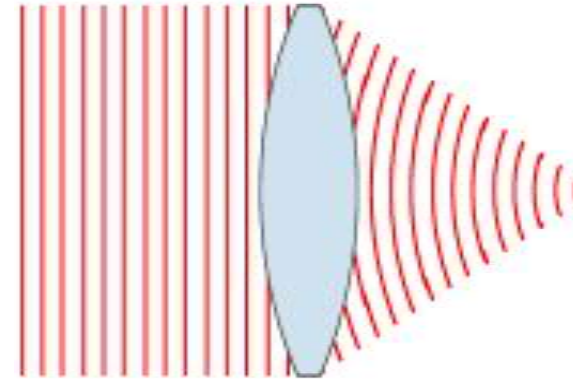


Point  $S$  and  $P$  are spoken of as conjugate points

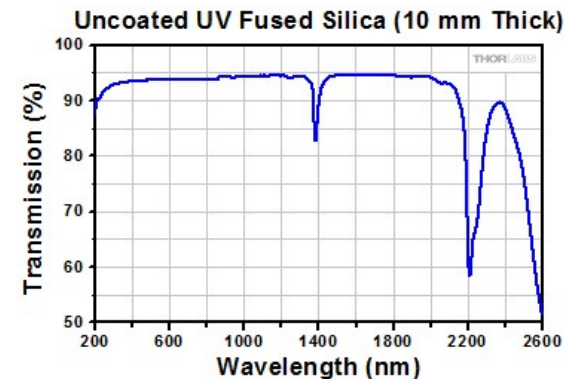
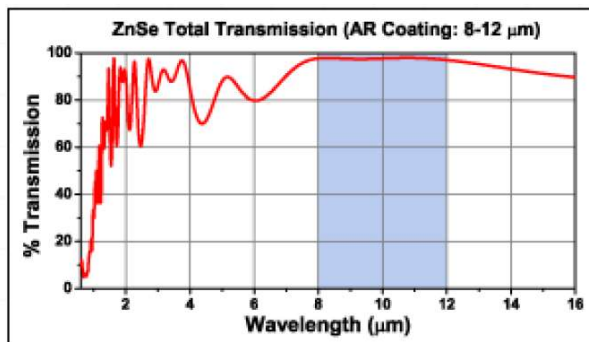
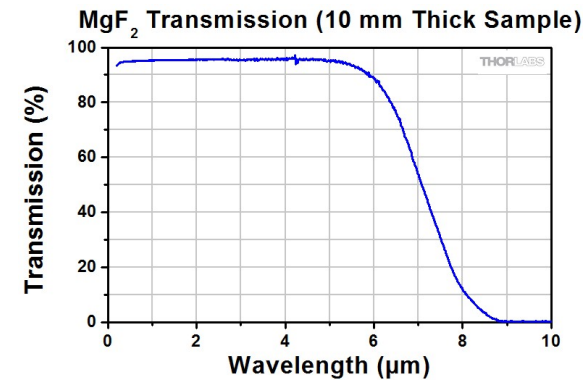
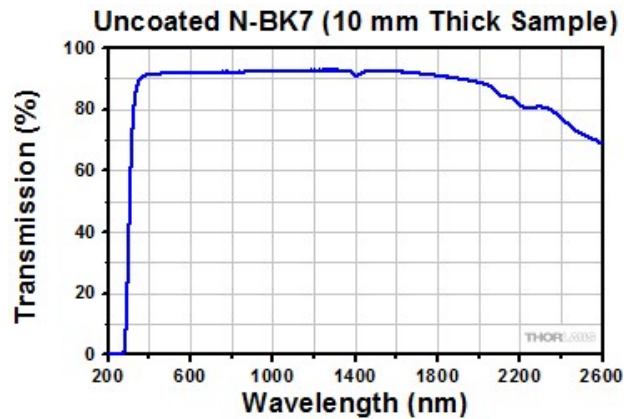
**Geometrical Optics:** as the wavelength of the radiant energy decreases in comparison to the physical dimensions of the optical system, the effects of diffraction become less significant. Point  $S$  and  $P$  are spoken of as conjugate points

## 4.2 Lenses

A lens is a refracting device that reconfigures a transmitted energy distribution.



For UV, lightwave, IR, microwave



## 4.2.1 Aspherical Surfaces

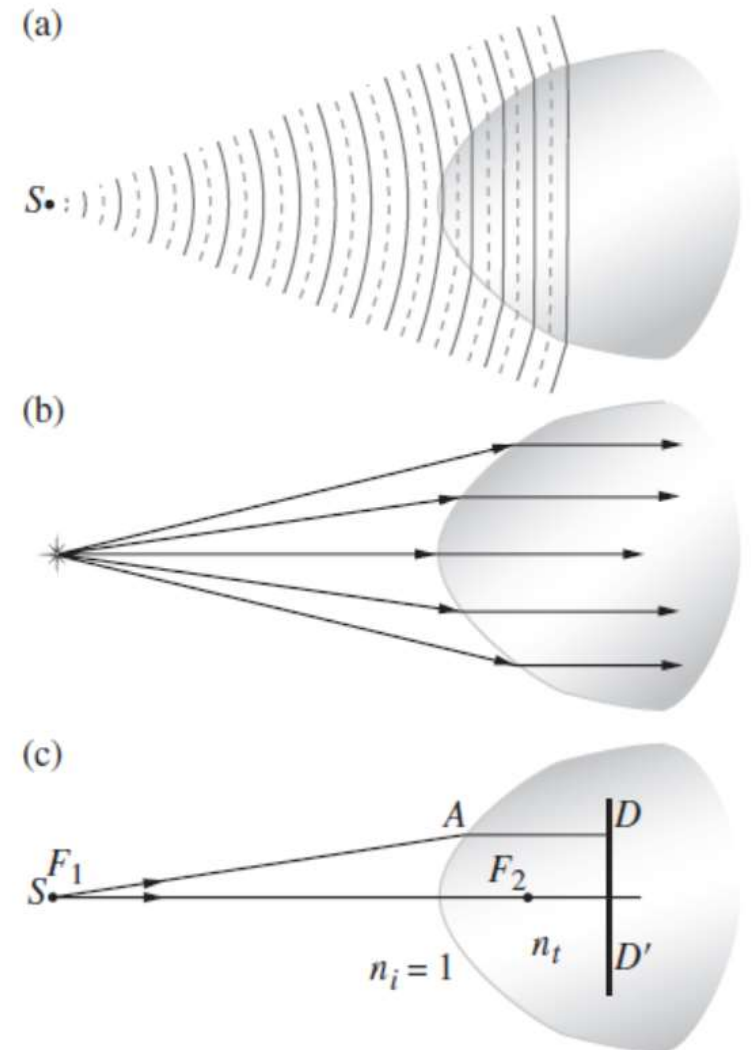
Whatever path the light takes from S to  $\overline{DD'}$ , it must always be the same number of wavelengths long, so that the disturbance begins and ends in-phase.

$$\overline{F_1A}/\lambda_i + \overline{AD}/\lambda_t = C_1$$

$$n_i(\overline{F_1A}) + n_t(\overline{AD}) = C_2$$

$$\overline{F_1A}/v_i + \overline{AD}/v_t = C_3$$

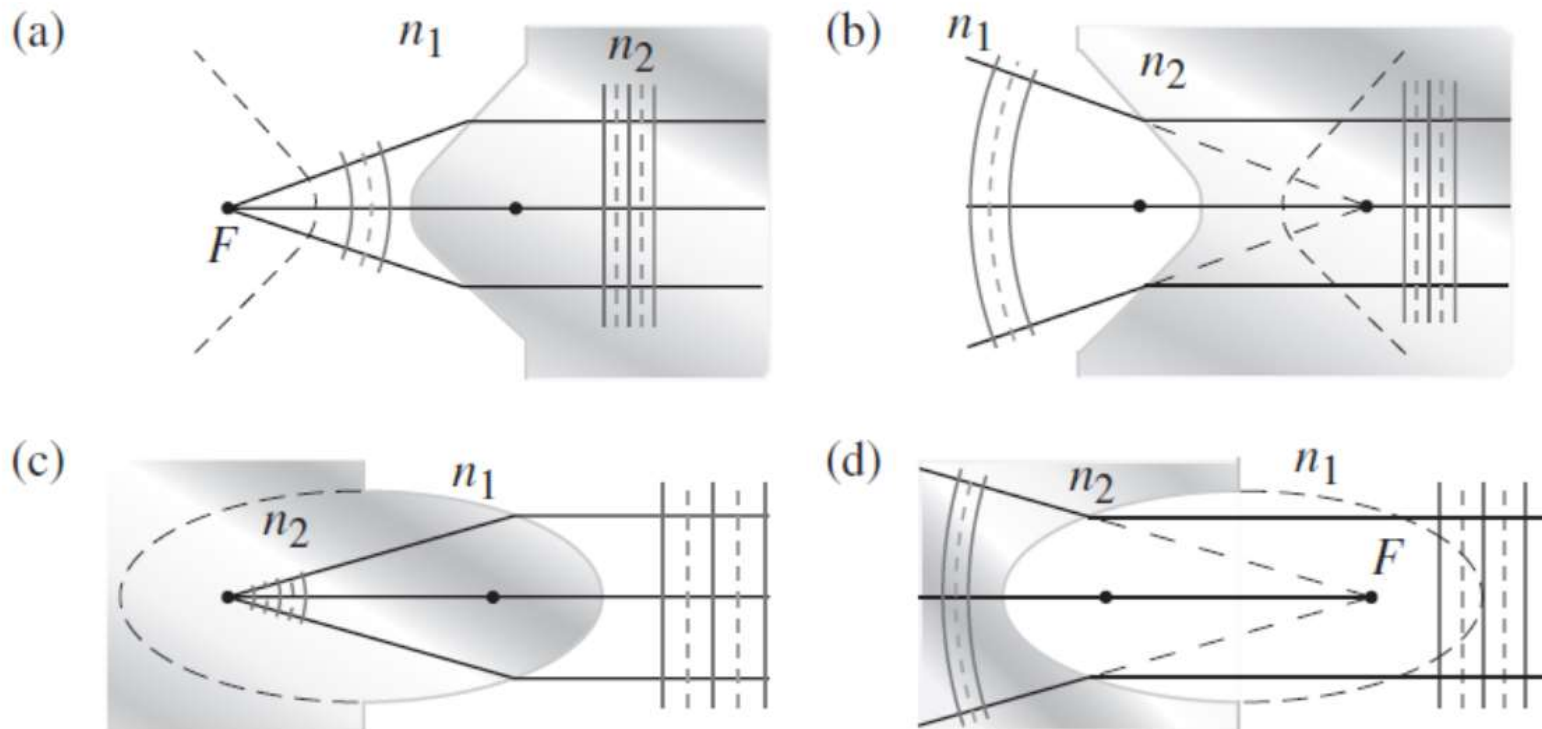
All paths from S to  $\overline{DD'}$  must take the same amount of time to traverse.



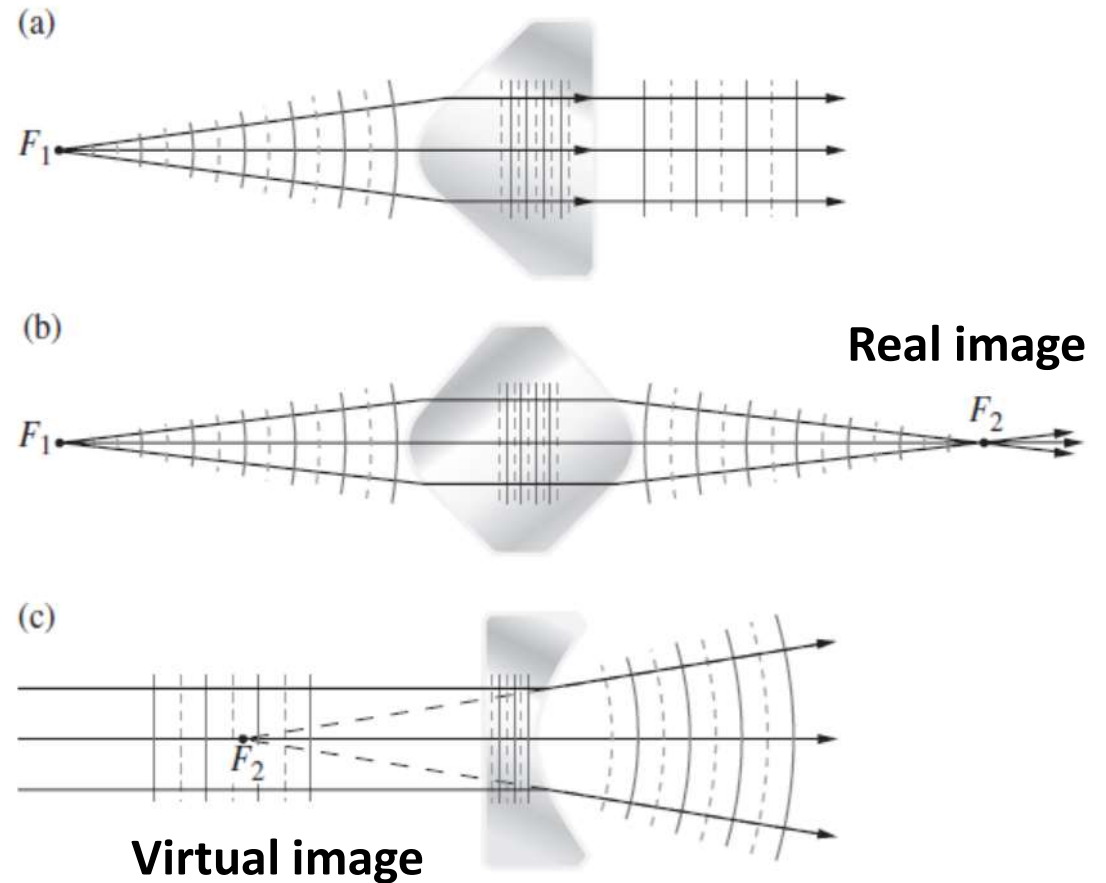
$$(\overline{F_1A}) + \frac{n_t}{n_i}(\overline{AD}) = \text{constant}$$

Eccentricity (偏心率):  $e = n_{ti} > 1$

The greater the eccentricity, the flatter the hyperbola (双曲线).



- ❑ **Convex:** the lenses are thicker at their midpoints than at their edges.
- ❑ **Converging lenses:** each lens causes the incoming beam to converge somewhat, to bend a bit more toward the central axis.
- ❑ **Concave:** the lenses are thinner in the middle than at the edges.
- ❑ **Diverging lenses:** these devices turn rays outward away from the central axis.



## 4.2.2 Refraction at Spherical Surfaces

顶点, 交点

Vertex: the point  $V$

Objective distance:  $s_o = \overline{SV}$

Image distance:  $s_i = \overline{VP}$

像距

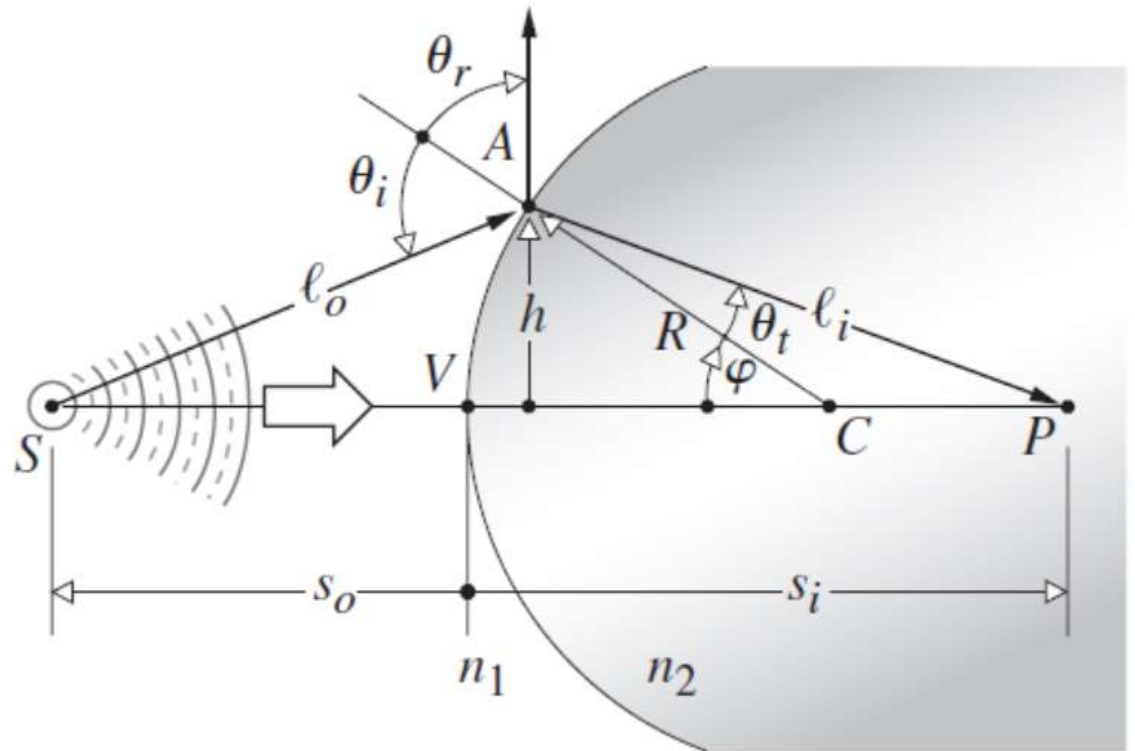
Fermat's Principle:

$$\frac{d(OPL)}{d\varphi} = 0$$

where  $OPL = n_1 \ell_o + n_2 \ell_i$

We have

$$\frac{n_1}{\ell_o} + \frac{n_2}{\ell_i} = \frac{1}{R} \left( \frac{n_2 s_i}{\ell_i} - \frac{n_1 s_o}{\ell_o} \right)$$





First-order theory:

$$\cos \varphi = 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \dots$$

$$\sin \varphi = \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \dots$$

$$\frac{n_1}{\ell_0} + \frac{n_2}{\ell_i} = \frac{1}{R} \left( \frac{n_2 s_i}{\ell_i} - \frac{n_1 s_0}{\ell_0} \right)$$

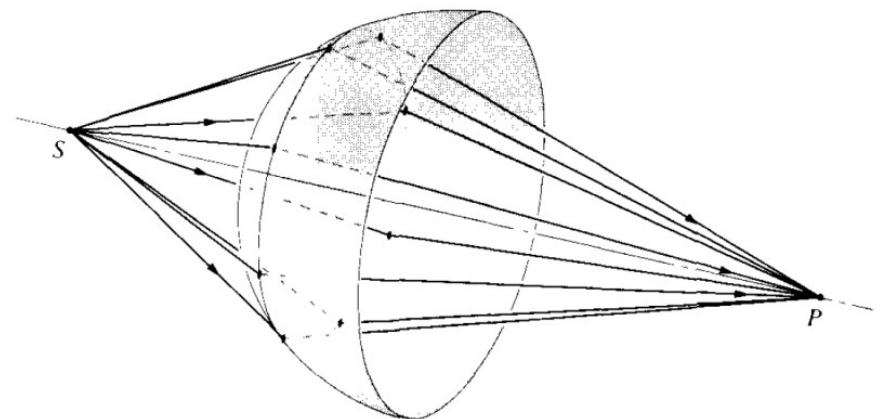


$$\frac{n_1}{s_0} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

**Paraxial (傍轴) rays:** rays that arrive at shallow angles with respect to the optical axis.

Same meaning: first-order, paraxial, or Gaussian optics

**Paraxial region:** the emerging wavefront segment corresponding to these paraxial rays is essentially spherical and will form a "perfect" image at its center  $p$  located at  $s_i$ .

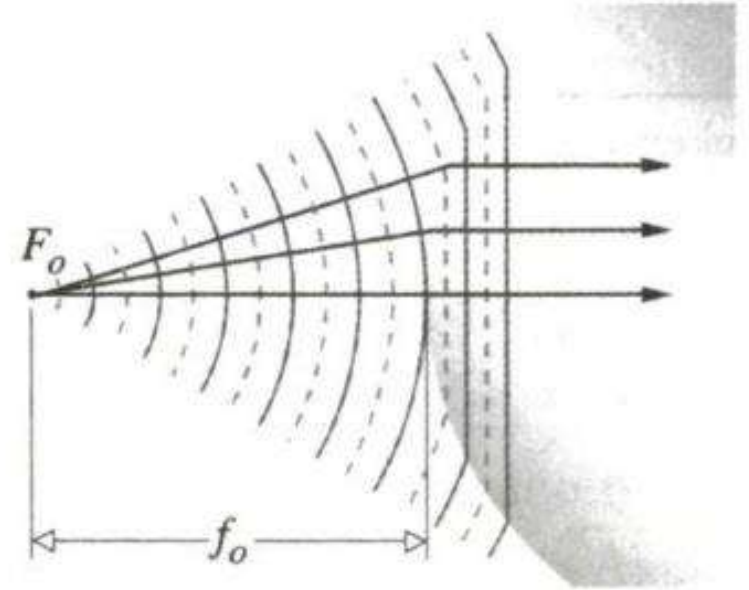


When  $s_i = \infty$

$$\frac{n_1}{s_o} + \frac{n_2}{\infty} = \frac{n_2 - n_1}{R}$$

First or objective focal length:  $s_o \equiv f_o$

$$f_o = \frac{n_1}{n_2 - n_1} R$$

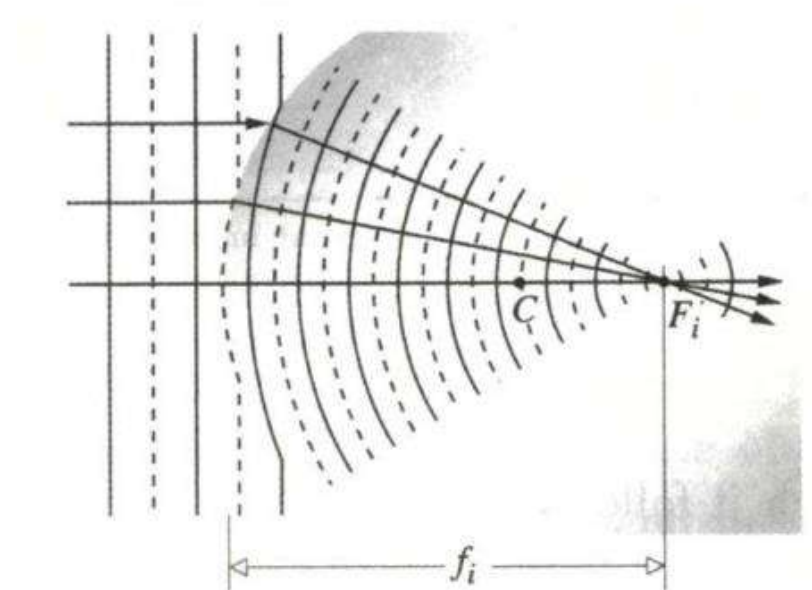


When  $s_o = \infty$

$$\frac{n_1}{\infty} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

Second or image focal length:  $s_i \equiv f_i$

$$f_i = \frac{n_2}{n_2 - n_1} R$$

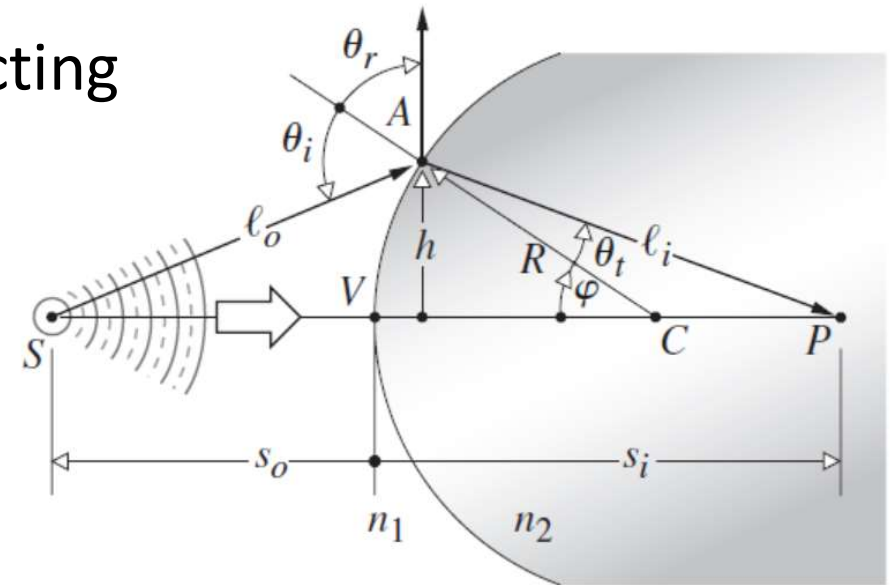


Sign convention for spherical refracting surfaces and thin lenses  
(light entering from the left)

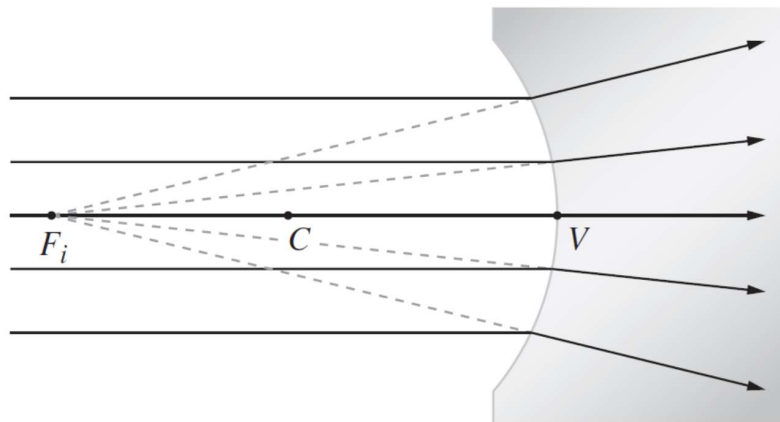
$s_o, f_o : +$  left of  $V$

$s_i, f_i : +$  right of  $V$

$R : +$  if  $C$  is right of  $V$

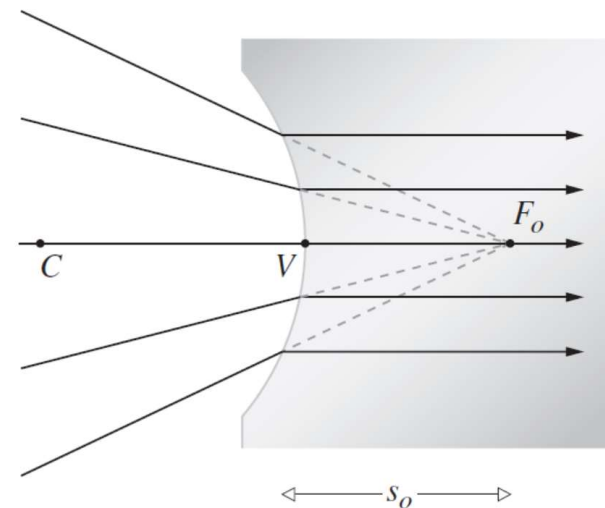


$R, f_i, s_i$ : negative



An image is virtual when the rays diverge from it.

$s_o, R, f_o, s_i$ : negative

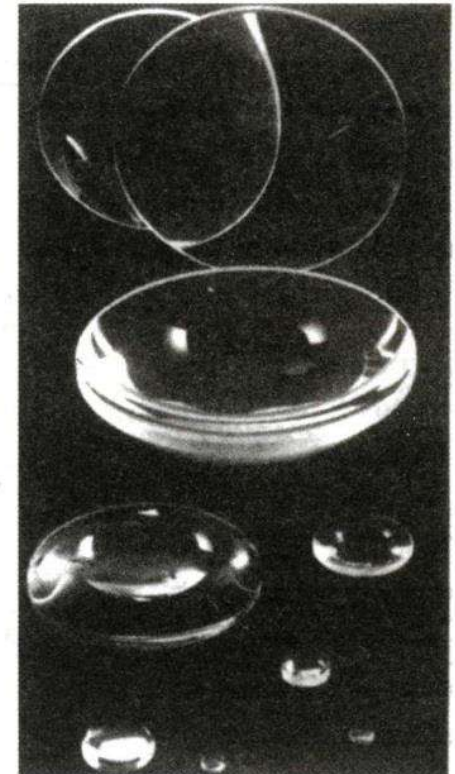


An object is virtual when the rays converge toward it.

## 4.2.3 Thin lenses

- Convex, converging, or positive lenses: thick at the center and so tend to decrease the radius of curvature of the wavefronts. (index of the lens is greater than that of the media)
- Concave, diverging, or negative lenses: thin at the center and tend to advance that portion of wavefront, causing it to diverge more than it did upon entry.

| CONVEX                                                                                                                        | CONCAVE                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|  $R_1 > 0$ $R_2 < 0$<br>Bi-convex          |  $R_1 < 0$ $R_2 > 0$<br>Bi-concave          |
|  $R_1 = \infty$ $R_2 < 0$<br>Planar-convex |  $R_1 = \infty$ $R_2 > 0$<br>Planar-concave |
|  $R_1 > 0$ $R_2 > 0$<br>Meniscus convex  |  $R_1 > 0$ $R_2 > 0$<br>Meniscus concave  |



In the paraxial region, the location of the conjugate points S and P is given by

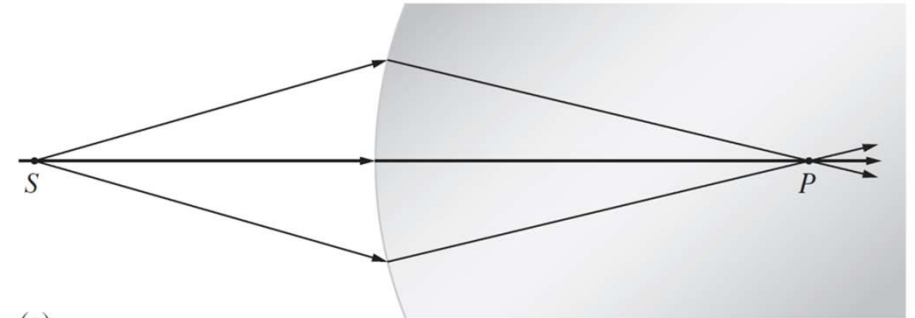
$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

As  $s_o$  decreases,  $s_i$  moves away from the vertex.

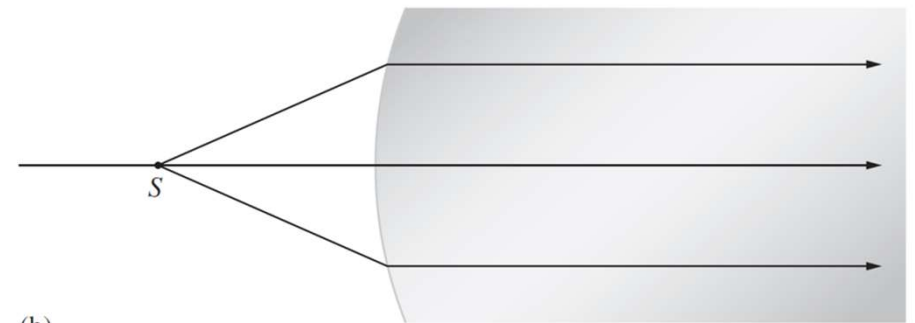
Finally,  $s_o = f_0$ ,  $s_i = \infty$ .

If  $s_o$  gets any smaller,  $s_i$  will have to be negative.

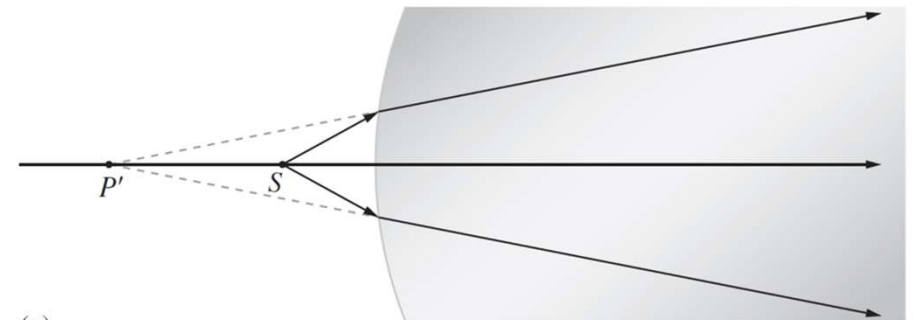
In other words, the image becomes virtual.



(a)



(b)



(c)

# Thin-lens equations

In the paraxial region

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

At the first surface, paraxial rays issuing from  $S$  at  $s_{o1}$  will meet at  $P'$ , a distance, which we now call  $s_{i1}$ , from  $V_1$ , given by

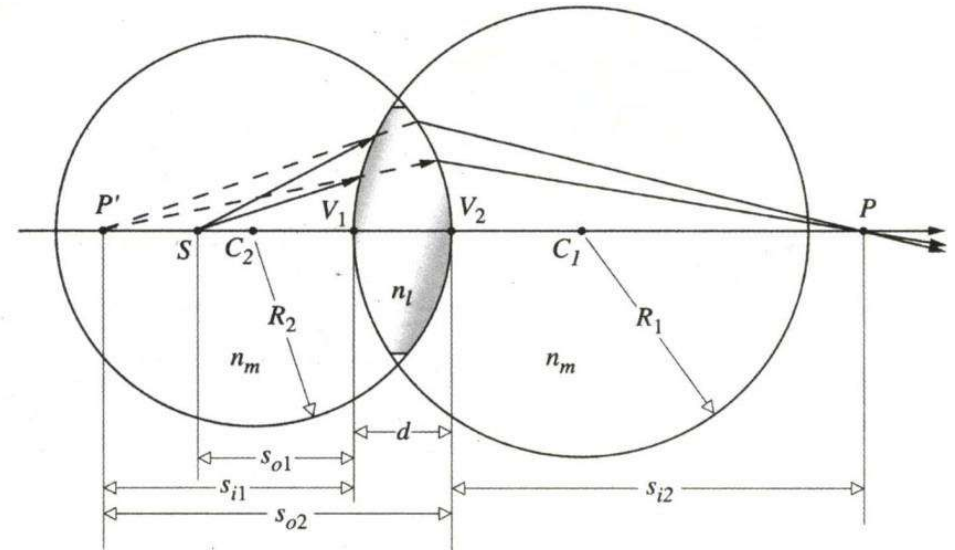
$$\frac{n_m}{s_{o1}} + \frac{n_l}{s_{i1}} = \frac{n_l - n_m}{R_1}$$

At the second surface:

$$\frac{n_l}{-s_{i1} + d} + \frac{n_m}{s_{i2}} = \frac{n_m - n_l}{R_2}$$

Add above equations, we have

$$\frac{n_m}{s_{o1}} + \frac{n_m}{s_{i2}} = (n_l - n_m) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{n_l d}{(s_{i1} - d)s_{i1}}$$



For thin lens ( $d \rightarrow 0$ ) surrounding by air ( $n_m \approx 1$ )

**Thin-lens equation**

$$\frac{1}{s_o} + \frac{1}{s_i} = (n_l - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

Thin-lens equation

$$\frac{1}{s_o} + \frac{1}{s_i} = (n_l - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

The image focal length  $f_i$

$$\lim_{s_o \rightarrow \infty} s_i = f_i = \frac{R_2 R_1}{(n_l - 1)(R_2 - R_1)}$$

The object focal length  $f_o$

$$\lim_{s_i \rightarrow \infty} s_o = f_o = \frac{R_2 R_1}{(n_l - 1)(R_2 - R_1)}$$

**Gaussian lens formula**

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

where

$$\frac{1}{f} = (n_l - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

**Example:** compute the focal length in air of a thin planar-convex lens having a radius of curvature of 50 mm and an index of 1.5.

On the planar surface:  $R_1 = \infty$ ,  $R_2 = -50$

$$\frac{1}{f} = (1.5 - 1) \left( \frac{1}{\infty} - \frac{1}{-50} \right), f = 100 \text{ mm}$$

On the curved surface:  $R_1 = 50$ ,  $R_2 = \infty$

$$\frac{1}{f} = (1.5 - 1) \left( \frac{1}{50} - \frac{1}{\infty} \right), f = 100 \text{ mm}$$

If an object is alternately placed at distances 600 mm, 200 mm, 150mm, 100mm, and 50 mm from the lens on either side, where can we find the image points.

$$s_i = \frac{s_o f}{s_o - f} = \frac{100 s_o}{s_o - 100}$$

$$s_i = 120, 200, 300, \infty, \text{ and } -100 \text{ mm}$$

# Optical Center

A beam entering a planar transparent plate emerges parallel to its initial direction

$$\theta_i = \theta_t$$

The lateral displacement of the beam

$$a = \frac{d \sin(\theta_i - \theta_t)}{\cos \theta_t}$$

A ray traversing a medium bounded by parallel planes will be displaced laterally but will suffer no angular deviation.

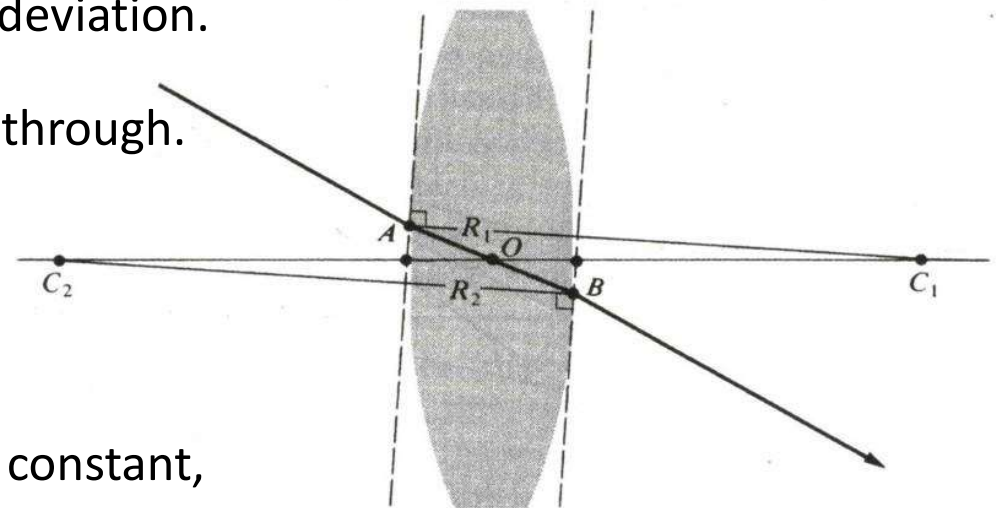
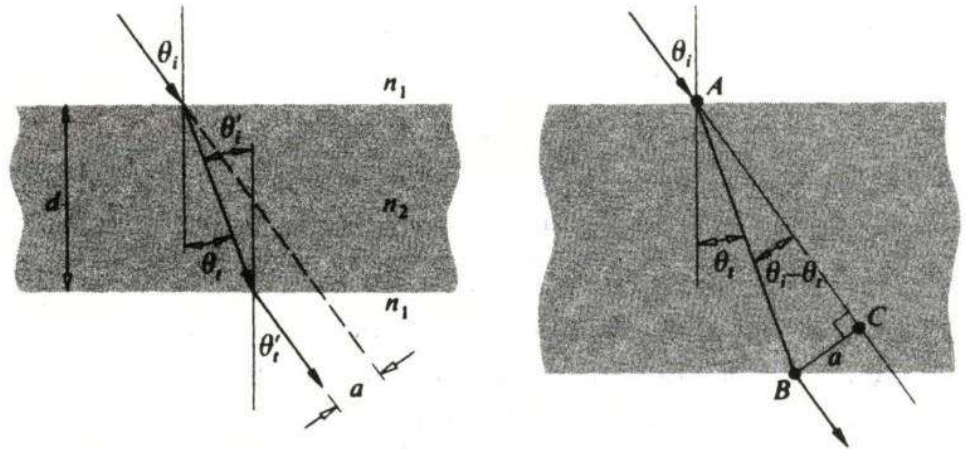
**Optical center:** the point that all such rays pass through.

Triangles  $AOC_1$  and  $BOC_2$  are similar

$$|R_1|(\overline{OC_2}) = |R_2|(\overline{OC_1})$$

Since the radii are constant, the location of O is constant, independent of A and B.

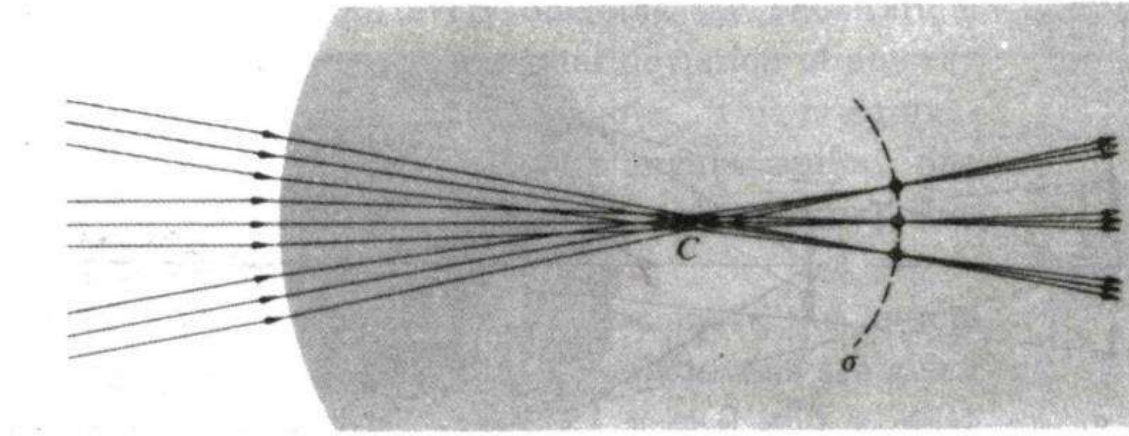
**For thin lens, rays passing through O may be drawn as straight lines.**





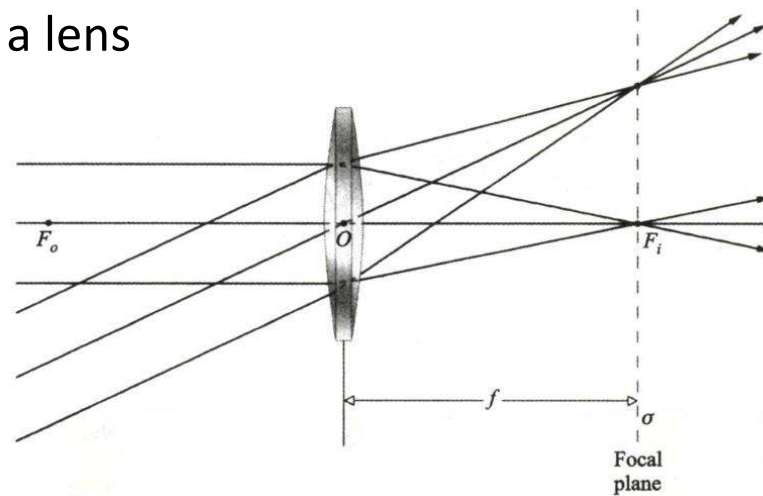
# Focal Points and Planes

Focal plane of a spherical refracting surface

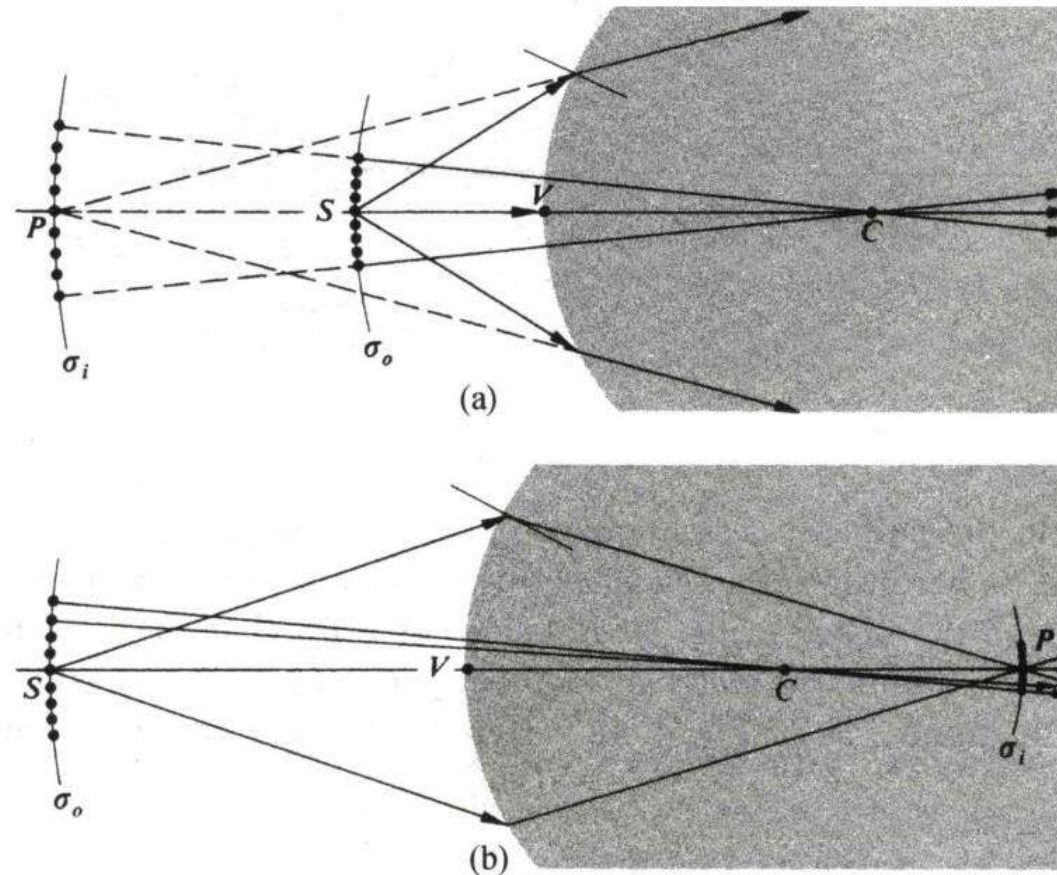


Since the ray cone must indeed be narrow,  $\sigma$  can be represented as a plane normal to the symmetry axis and passing through the image focus.

Similarly, focal plane of a lens



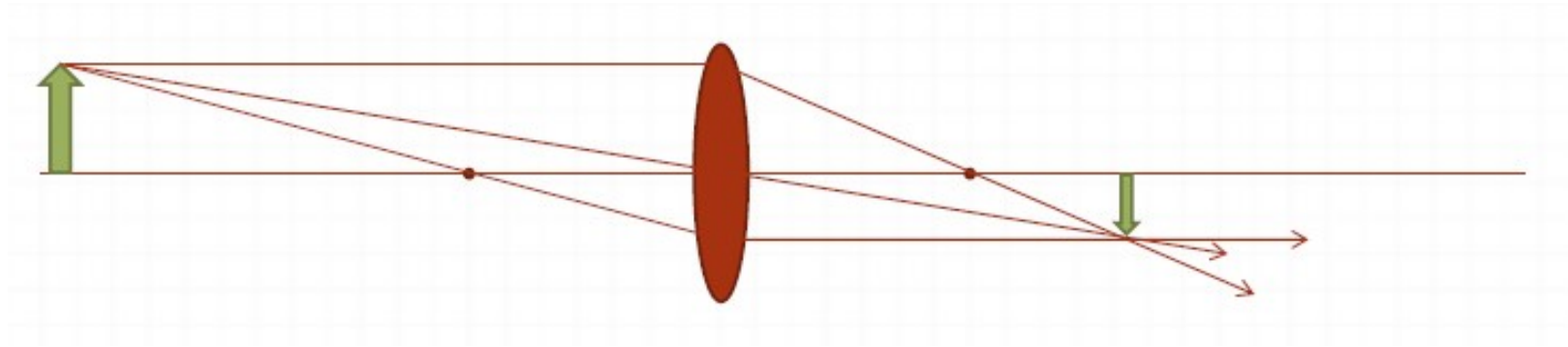
## Finite imagery



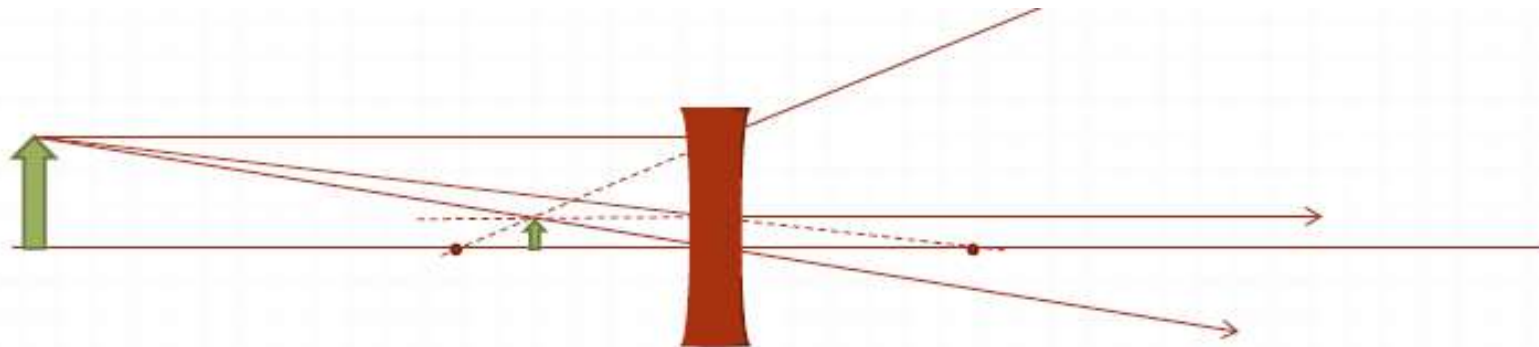
The final image formed by a lens of a small planar object normal to the optical axis will itself be a small plane normal to that axis

## Ray diagrams

A real object and a positive lens



A real object and a negative lens



Triangles  $AOF_i$  and  $P_2P_1F_i$  are similar.

$$\frac{y_o}{|y_i|} = \frac{f}{s_i - f}$$

Triangles  $S_2S_1O$  and  $P_2P_1O$  are similar.

$$\frac{y_o}{|y_i|} = \frac{s_o}{s_i}$$

So  $\frac{s_o}{s_i} = \frac{f}{s_i - f}$

Gaussian lens formula

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

Triangles  $S_2S_1F_o$  and  $BOF_o$  are similar.

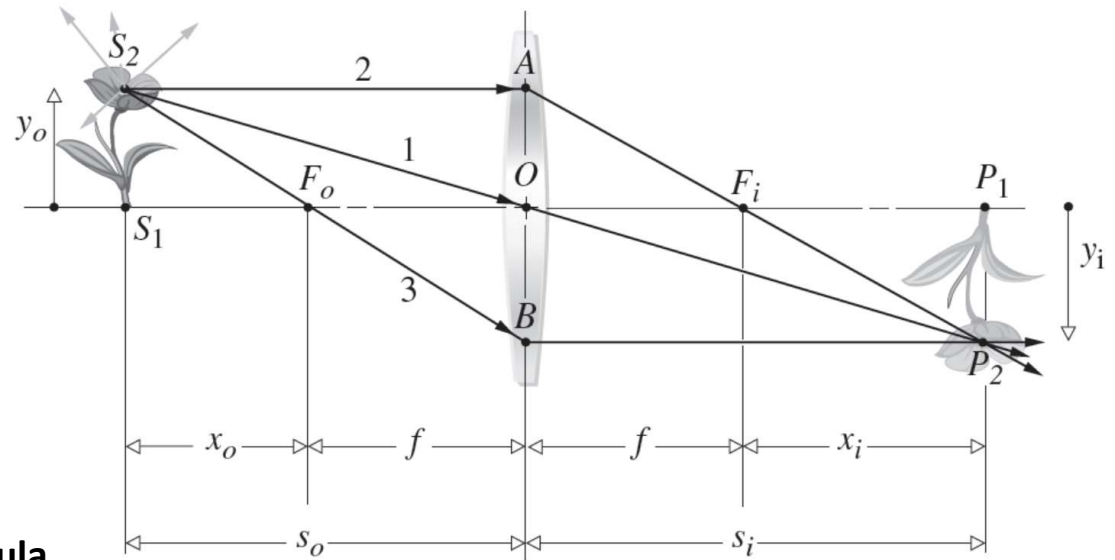
$$\frac{f}{s_o - f} = \frac{|y_i|}{y_o}$$

$$s_o - f = x_o \quad s_i - f = x_i$$

**Newtonian form** of the lens equation

$$x_o x_i = f^2$$

The object and image must be on opposite sides of their respective focal points



Sign convention for spherical refracting surfaces and thin lenses  
(light entering from the left)

$s_o, f_o$  : + left of  $V$

$x_o$  : + left of  $F_o$

$s_i, f_i$  : + right of  $V$

$x_i$  : + right of  $F_i$

$R$  : + if  $C$  is right of  $V$

$y_o, y_i$  : + above optical axis

Transverse magnification

$$M_T = \frac{y_i}{y_o}$$

A positive  $M_T$  connotes an erect image, while a negative value means the image is inverted.

$s_i$  and  $s_o$  are both positive for real objects and images, then all real images formed by a single thin lens will be inverted.

$$M_T = -\frac{s_i}{s_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

Thin-lens equation

As the object approaches the lens, the real image moves away from it.

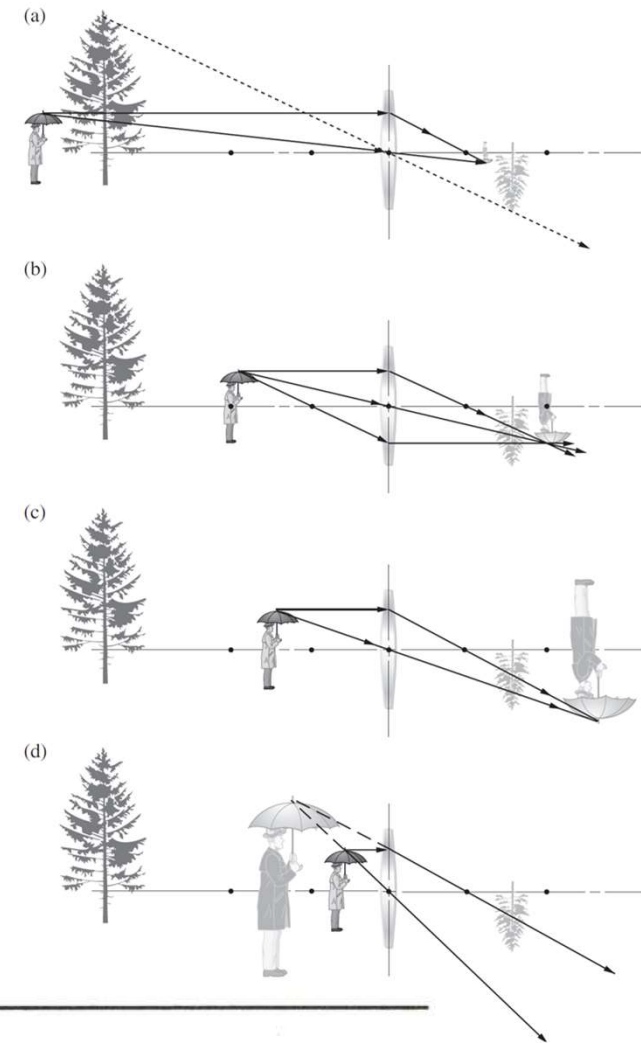
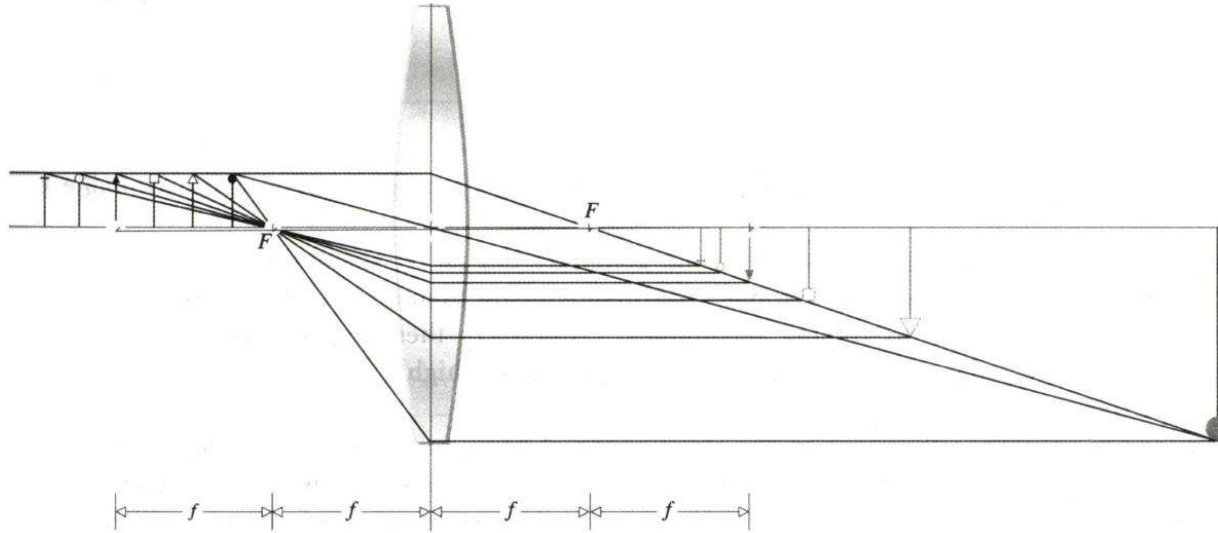


TABLE 4.3 Images of Real Objects Formed by Thin Lenses

| Convex              |         |                     |             |               |
|---------------------|---------|---------------------|-------------|---------------|
| Object              | Image   |                     |             |               |
| Location            | Type    | Location            | Orientation | Relative Size |
| $\infty > s_o > 2f$ | Real    | $f < s_i < 2f$      | Inverted    | Minified      |
| $s_o = 2f$          | Real    | $s_i = 2f$          | Inverted    | Same size     |
| $f < s_o < 2f$      | Real    | $\infty > s_i > 2f$ | Inverted    | Magnified     |
| $s_o = f$           |         | $\pm \infty$        |             |               |
| $s_o < f$           | Virtual | $ s_i  > s_o$       | Erect       | Magnified     |

The transformation from object to image space is not linear:

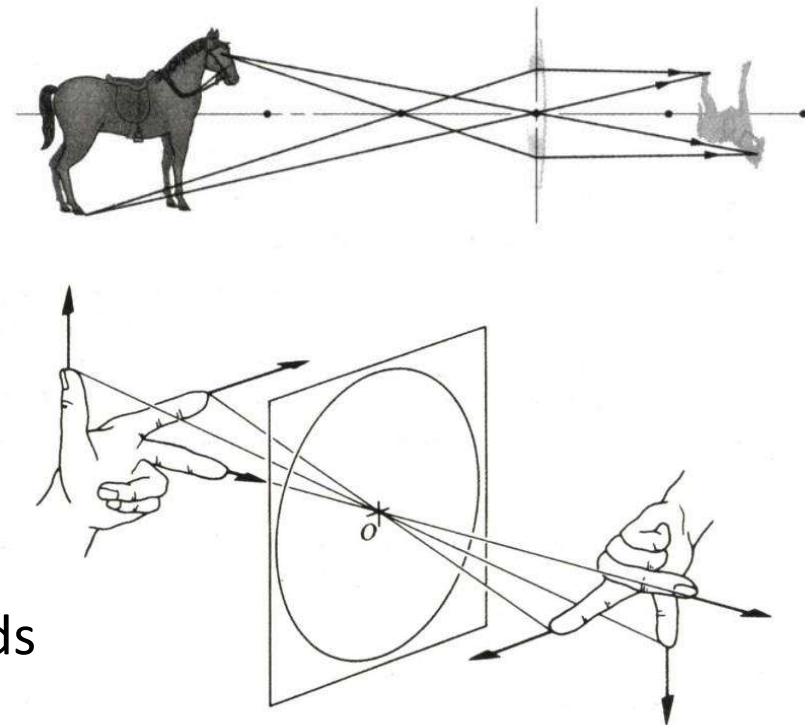


### Longitudinal magnification

$$\underbrace{M_L}_{\text{red wavy}} \equiv \frac{dx_i}{dx_o} = -\frac{f^2}{x_o^2} = -\underbrace{M_T^2}_{\text{red wavy}}$$

By differentiating  $x_o x_i = f^2$   $M_L = -M_T^2$

Evidently,  $M_L < 0$ : a positive  $dx_o$  corresponds to a negative  $dx_i$  and vice versa.

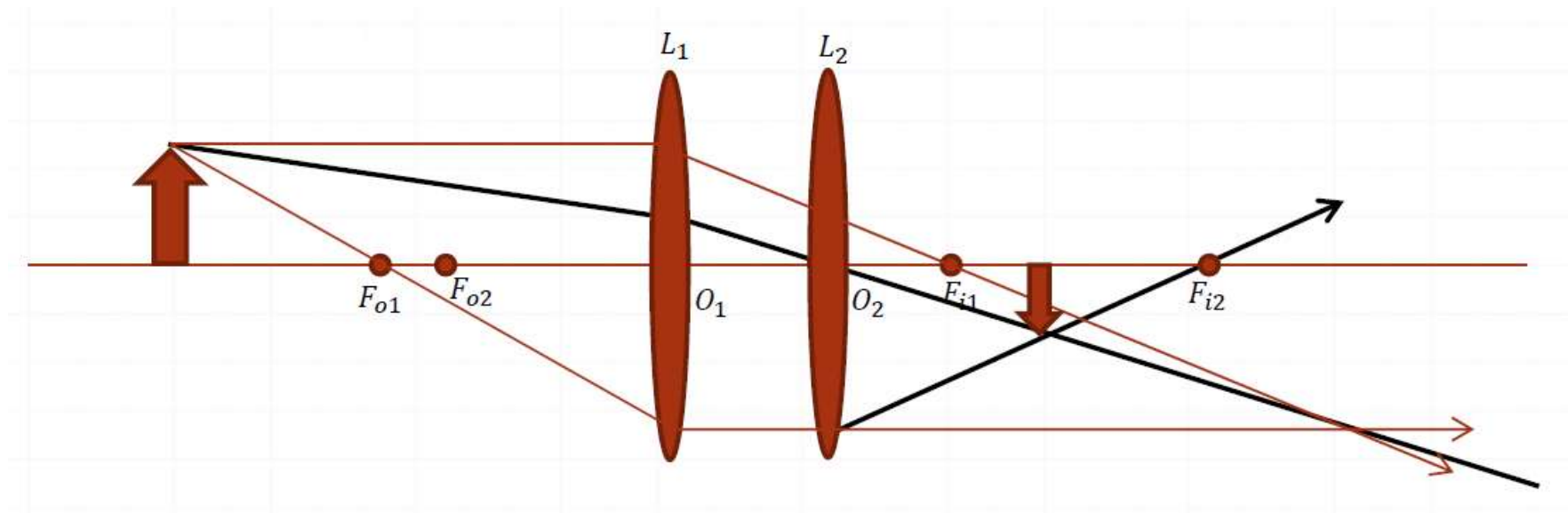




# Thin-Lens Combinations

Consider two thin positive lenses  $L_1$  and  $L_2$  separated by a distance  $d$ , which is smaller than either focal length.

The resulting image can be located graphically as follows

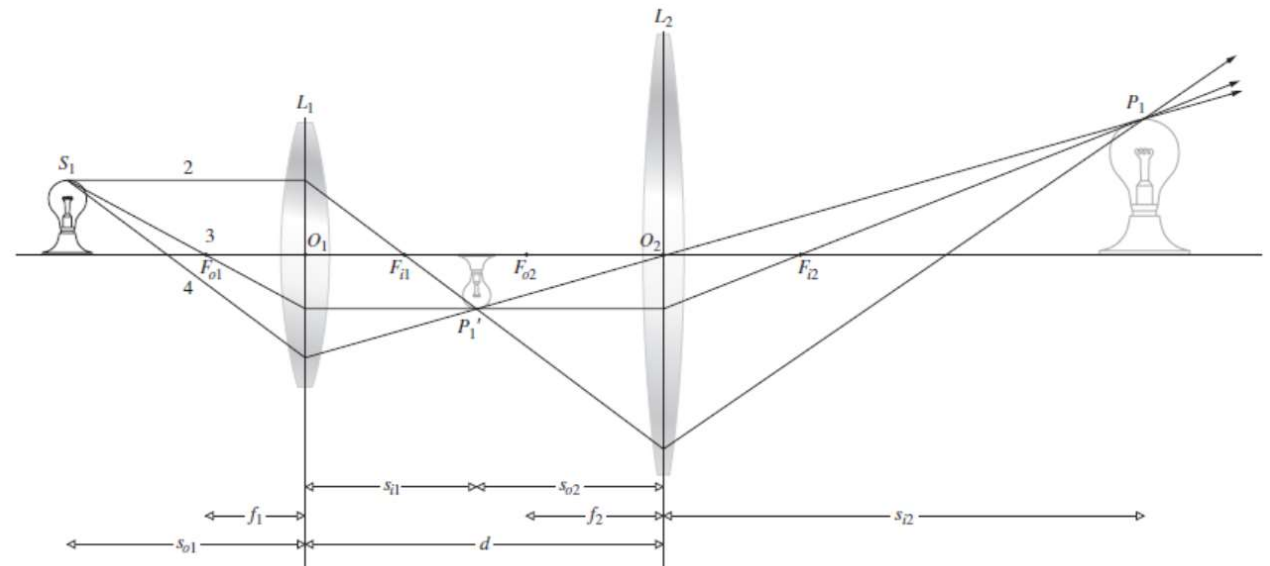


Consider a similar pair of lenses (increased separation)

$$\text{For } L_1 \quad \frac{1}{s_{i1}} = \frac{1}{f_1} - \frac{1}{s_{o1}}$$

$$\text{For } L_2 \quad \frac{1}{s_{i2}} = \frac{1}{f_2} - \frac{1}{s_{o2}}$$

$$s_{o2} = d - s_{i1}$$



If  $d > s_{i1}$ , the object for  $L_2$  is real.

If  $d < s_{i1}$ , the object for  $L_2$  is virtual.

Example: compute the image distance associated with an object place 50 cm from the first of two positive lenses. These in turn are separated by 20 cm and have focal lengths of 30 cm and 50 cm, respectively.

$$s_{i2} = \frac{f_2 d - f_2 s_{o1} f_1 / (s_{o1} - f_1)}{d - f_2 - s_{o1} f_1 / (s_{o1} - f_1)}$$

$$s_{i2} = \frac{50(20) - 50(50)(30)/(50 - 30)}{20 - 50 - 50(30)/(50 - 30)} = 26.2\text{cm}$$

$$M_T = M_{T1} M_{T2} = \frac{f_1 s_{i2}}{d(s_{o1} - f_1) - s_{o1} f_1}$$



## Front focal length

If we set  $s_{i2} \rightarrow \infty$ ,  $s_{o2}$  approaches  $f_2$

So  $s_{i1} \rightarrow d - f_2$

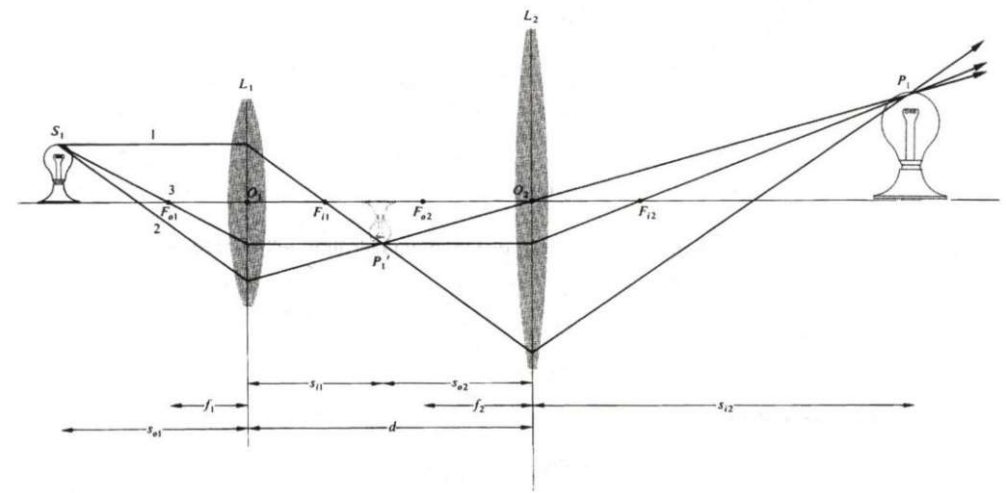
$$\left. \frac{1}{s_{o1}} \right|_{s_{i2} \rightarrow \infty} = \frac{1}{f_1} - \frac{1}{(d - f_2)} = \frac{d - (f_1 + f_2)}{f_1(d - f_2)}$$

$$\text{f.f.l.} = \frac{f_1(d - f_2)}{d - (f_1 + f_2)}$$

## Back focal length

If we set  $s_{o1} \rightarrow \infty$ ,  $(s_{o1} - f_1) \rightarrow s_{o1}$

$$s_{i2} = \frac{f_2 d - f_2 s_{o1} f_1 / (s_{o1} - f_1)}{d - f_2 - s_{o1} f_1 / (s_{o1} - f_1)} = \frac{f_2(d - f_1)}{d - (f_1 + f_2)} = \text{b.f.l.}$$



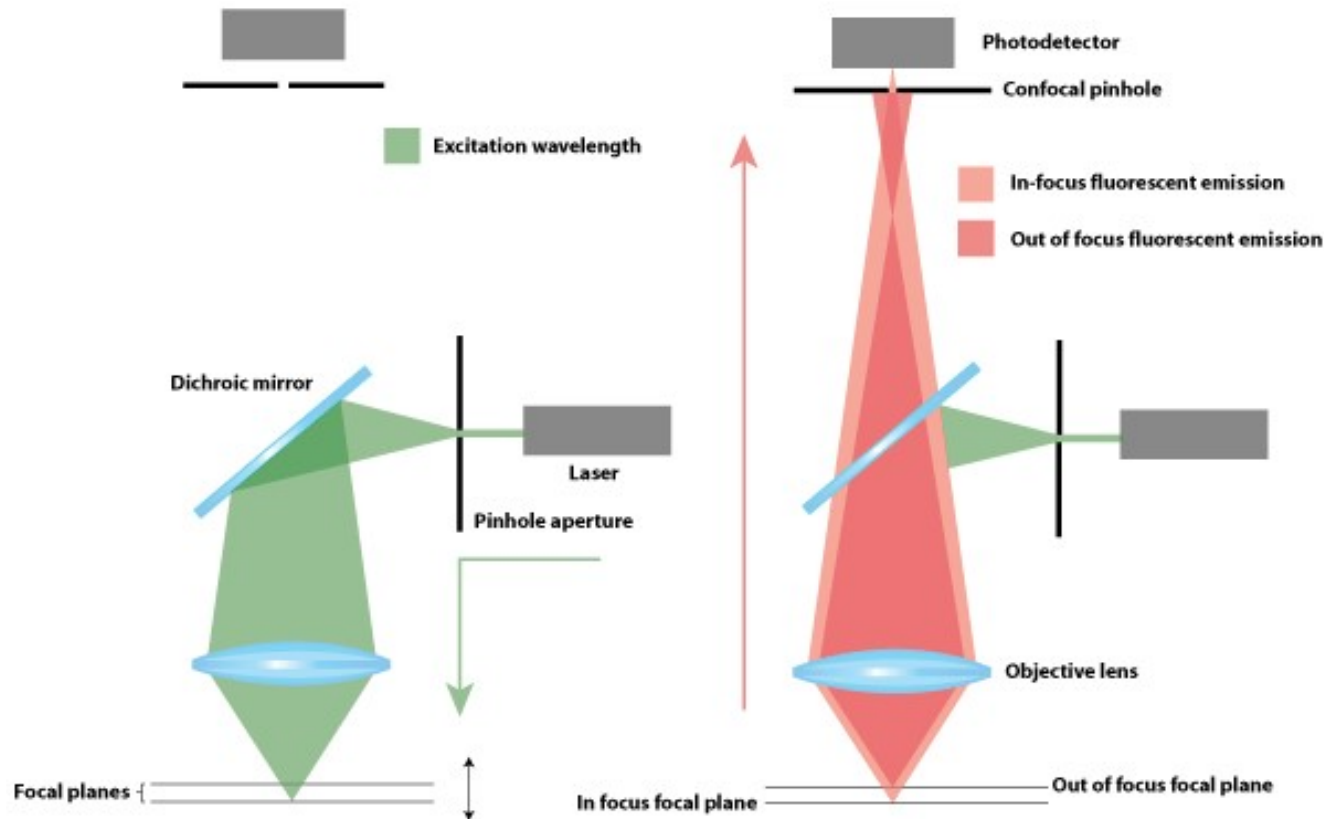
For  $d \rightarrow 0$

$$\text{f.f.l.} = \text{b.f.l.} = \frac{f_1 f_2}{f_1 + f_2}$$

Effective focal length

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

# Confocal microscopy



- ❑ The confocal microscope is so called because the illumination pinhole, the plane of focus within the specimen and the detector pinhole are all situated at conjugate focal planes. By restricting the illumination to a pinhole, rather than illuminating the entire field of view at once it becomes possible to build up a blur-free image in three dimensions.

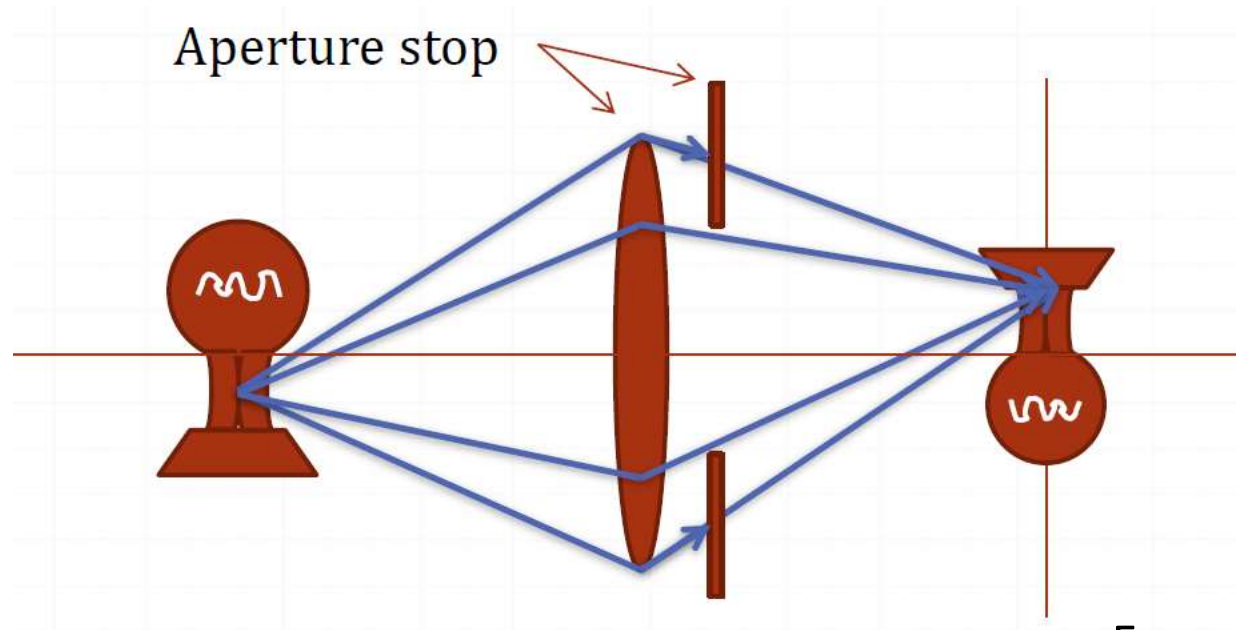
# Homework

Problems: 5.8, 5.13, 5.15, 5.24, 5.35, 5.37

# 4.3 Stops (光阑)

## Aperture and Field Stops

The intrinsically finite nature of all lenses demands that they collect only a fraction of the energy emitted by a point source.



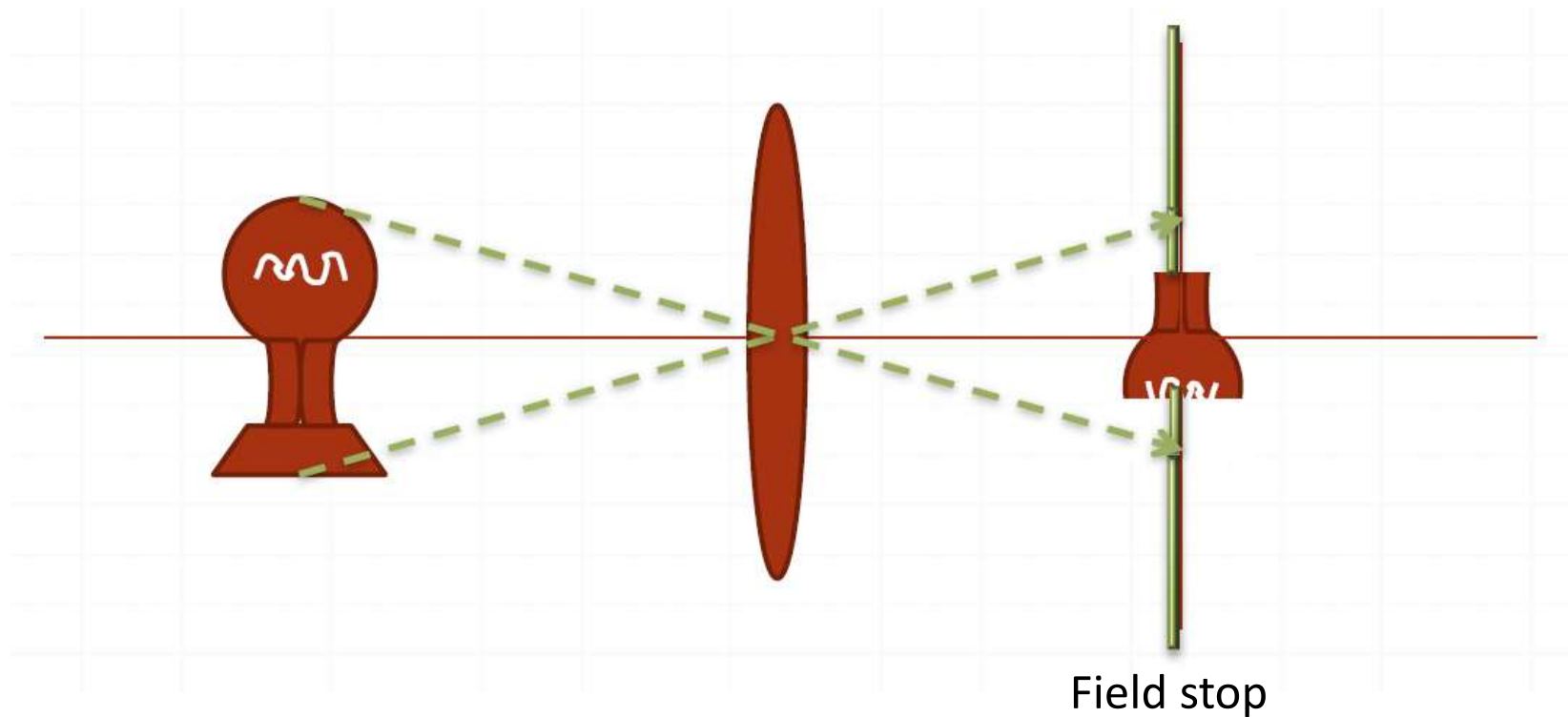
**Aperture stop (A.S., 孔径光阑):** any element, be it the rim of a lens or a separate diaphragm, that determines the amount of light reaching the image.

Example:  
diameter of the lens



The adjustable  
leaf diaphragm





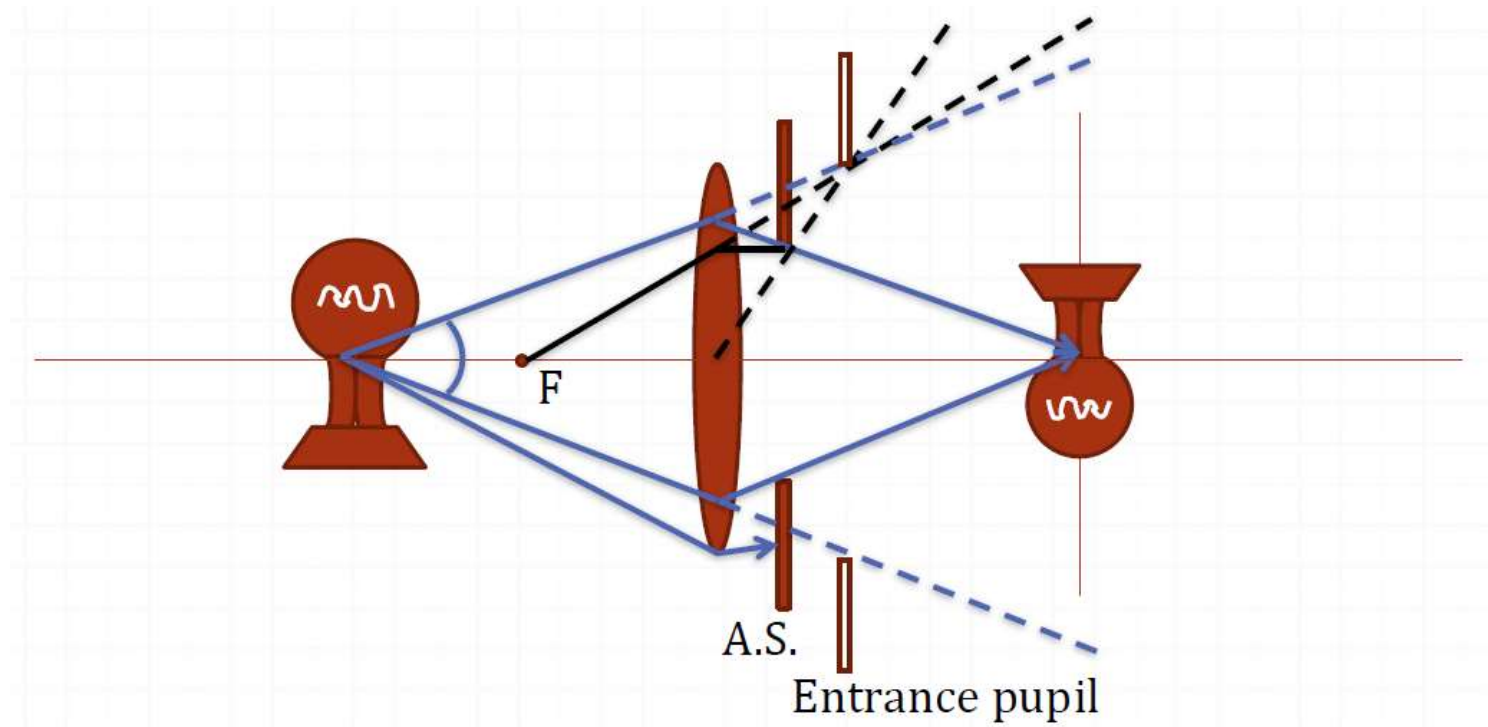
**Field stop (F. S., 视场光阑):** the element limiting the size or angular breadth of the object that can be imaged by the system. It determines the field of view of the instrument.

Example: the edge of CCD or film



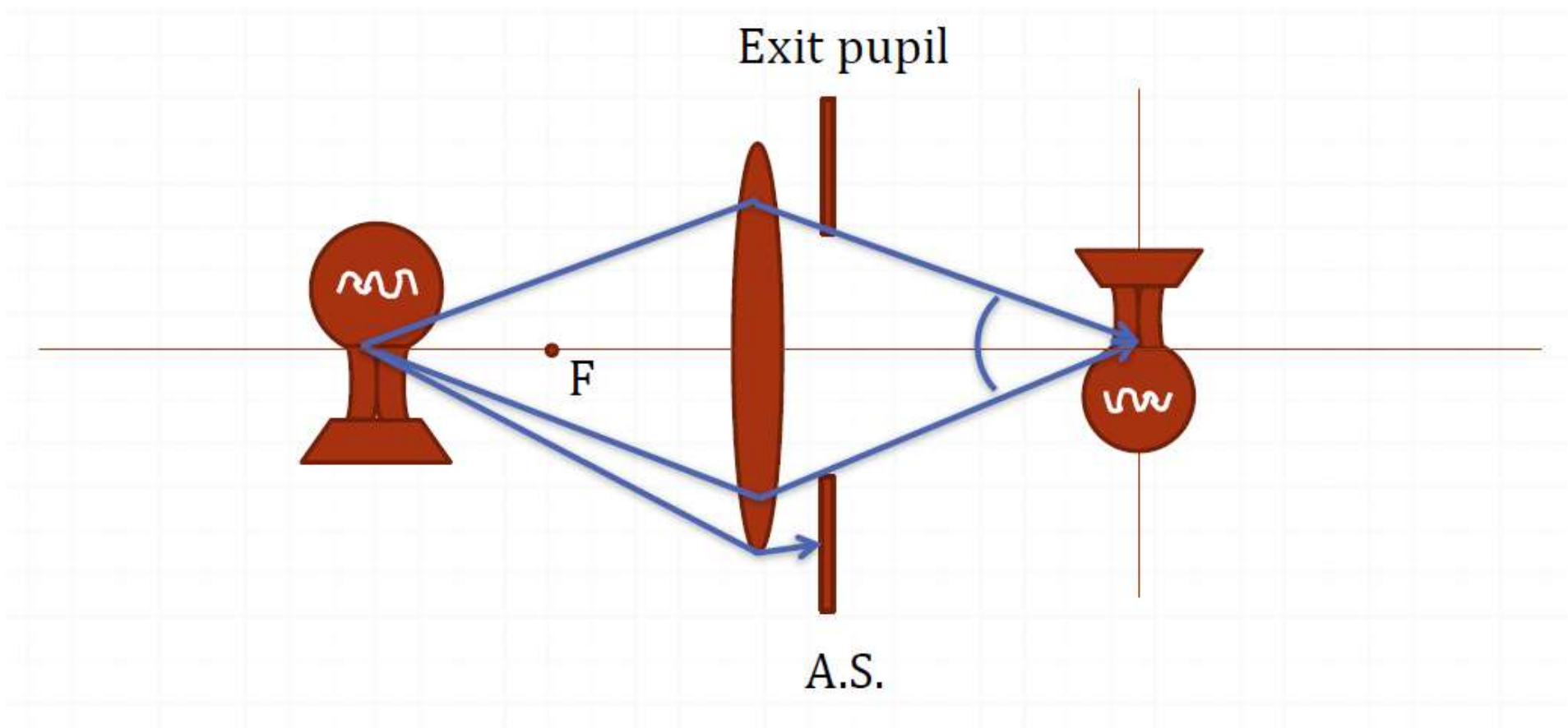
# Pupils (光瞳)

**Pupil:** determine whether or not a given ray will traverse the entire optical system, namely an image of the aperture stop.



**Entrance pupil of a system:** the image of the aperture stop as seen from an axial point on the object through those elements preceding the stop.

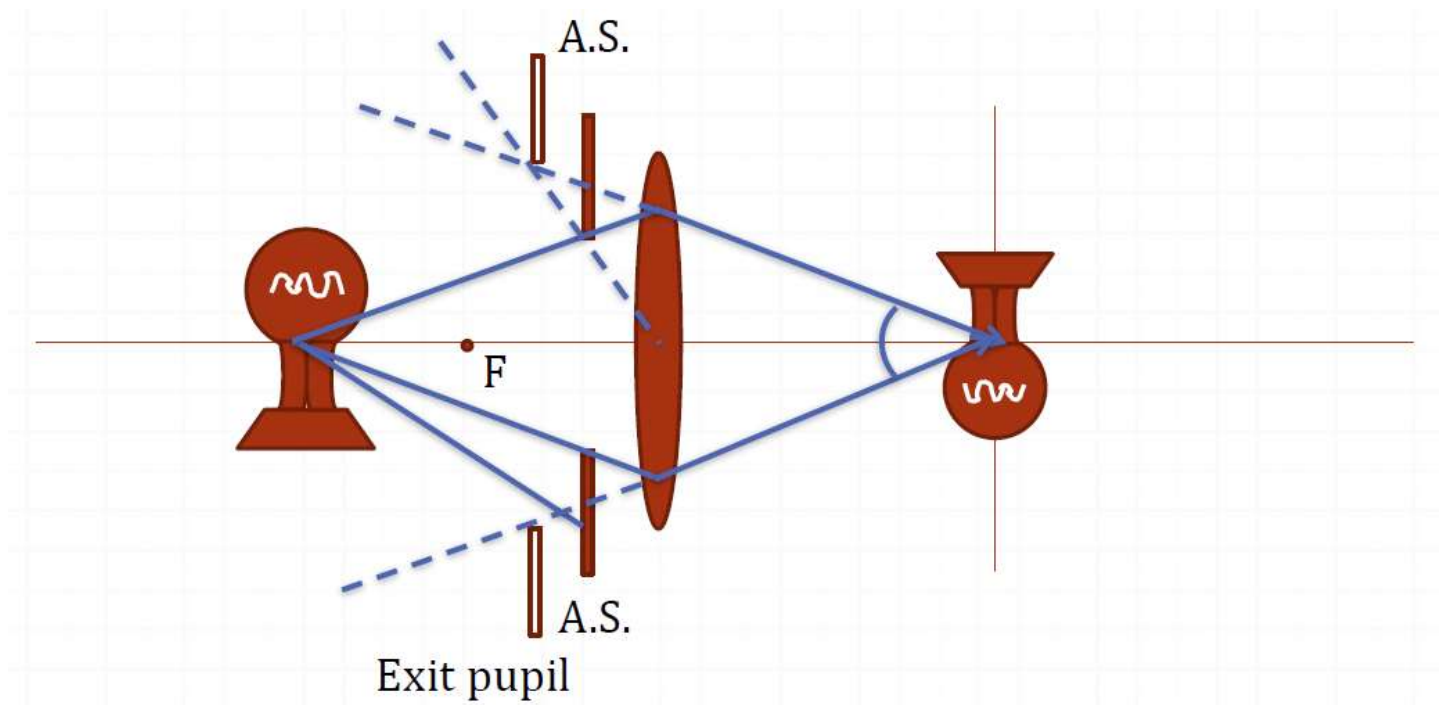
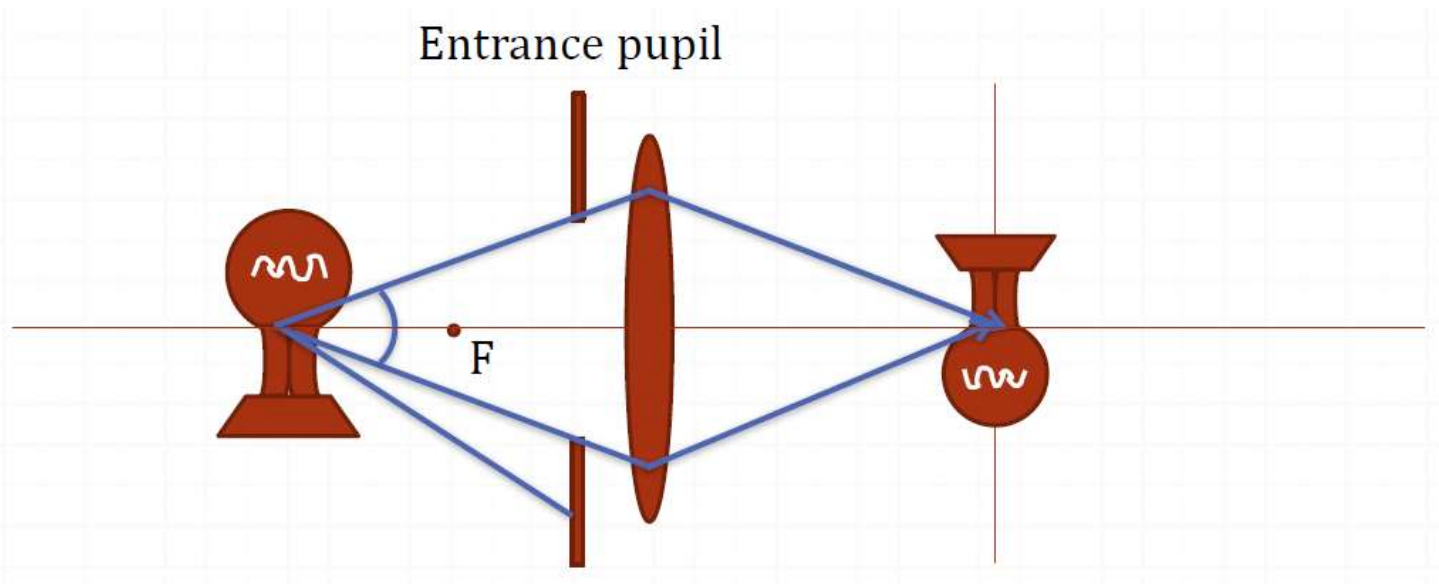
**Function:** the cone of light actually entering the optical system is determined by the entrance pupil.



**Exit pupil:** the image of the A. S. as seen from an axial point on the image plane through the interposed lenses, if there are any.

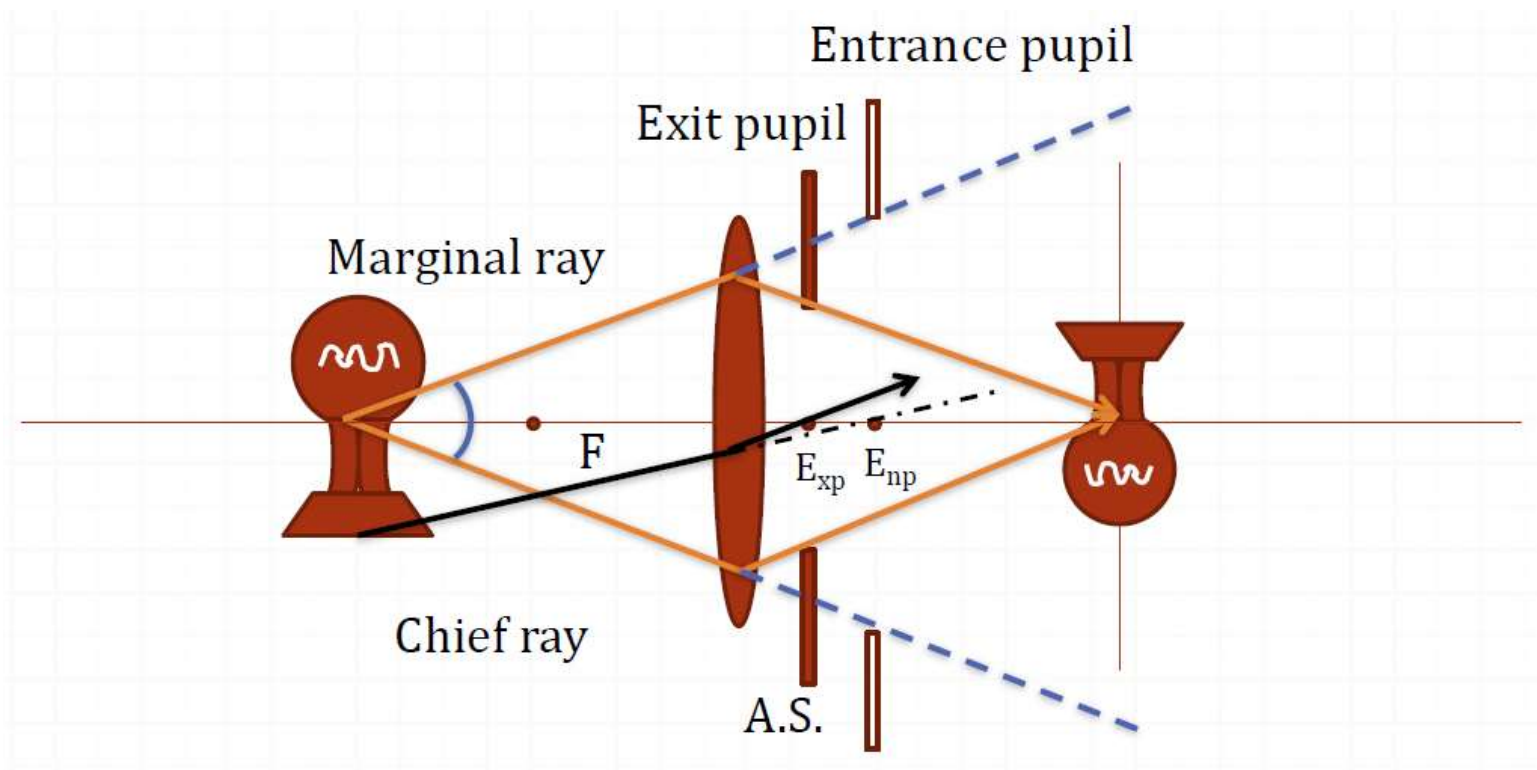
**Function:** the cone leaving it is controlled by the exit pupil.

Example: a front aperture stop F





# Chief and Marginal Ray



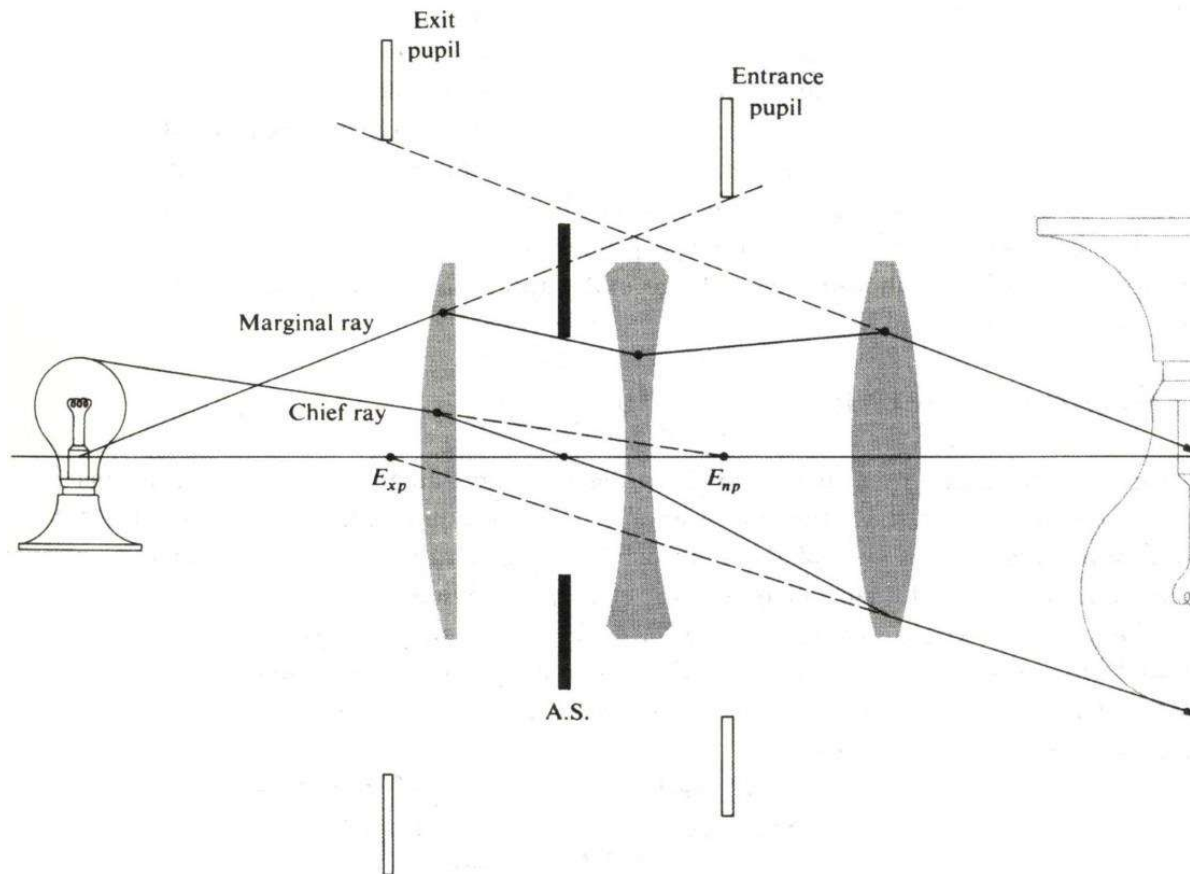
**Chief ray:** any ray from an off-axis object point that passes through the center of the aperture stop.

The chief ray enters the optical system along a line directed toward the midpoint of the entrance pupil,  $E_{np}$ . And leaves the system along a line passing through the center of the exit pupil,  $E_{xp}$ .

**Marginal ray:** it goes from the axial object point to the rim or margin of the entrance pupil.

# Entrance pupil for optical system

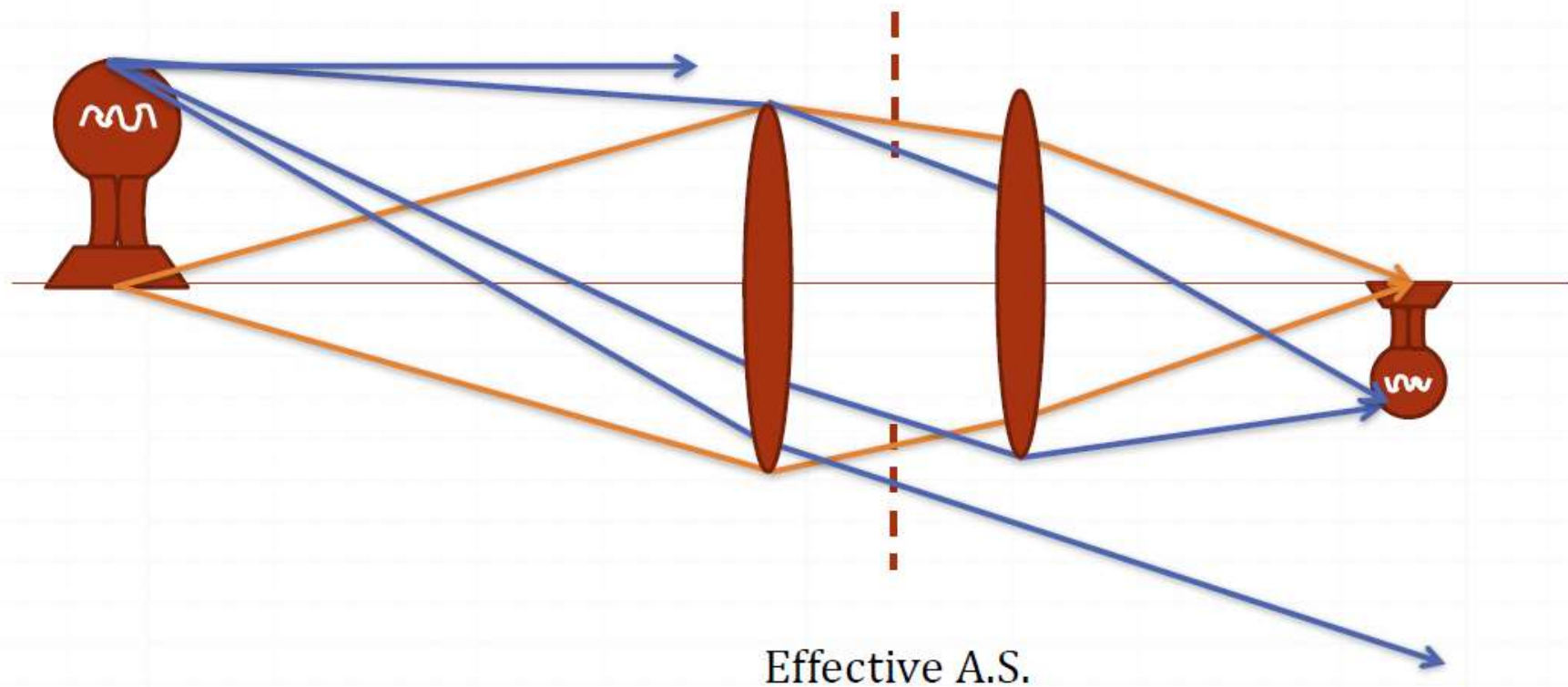
For a complex system, each component of the system must be imaged by the remaining elements to its left. The image that subtends the smallest angle at the axial object point is the **entrance pupil**.



Pupils and stops for a three-lens system

## Vignetting (渐晕效应)

The cone of rays that can reach the image plane becomes narrower due to the effective aperture stop, then the image is gradually faded out at points near its periphery.



## Relative Aperture (相对孔径)

**Irradiance** or flux density (the energy per unit area per unit time)

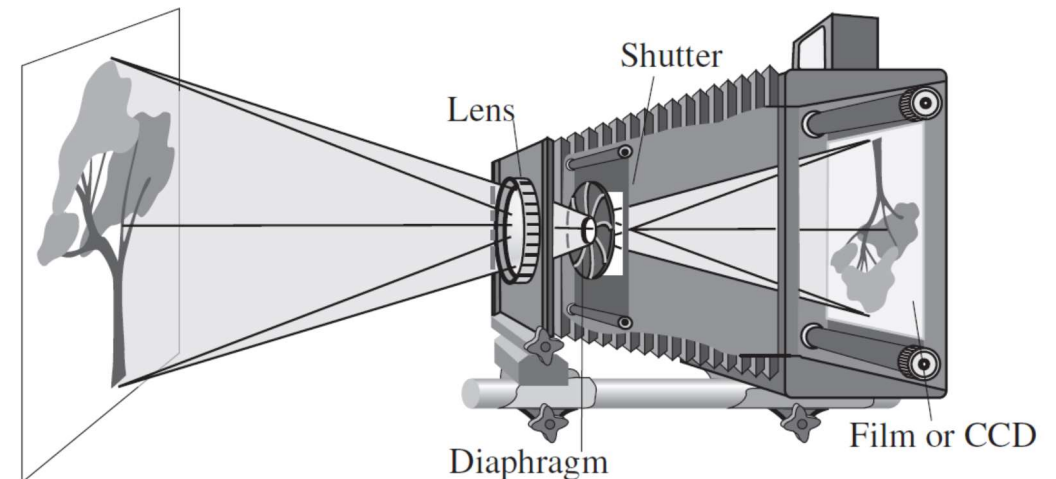
Proportional to **area of the entrance pupil**

Inversely proportional to **image area**

The entrance pupil area  $\propto$  square of its diameter ( $D^2$ )

$$\text{Since } M_T \equiv \frac{|y_i|}{y_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$

The image area  $\propto$  square of focus ( $f^2$ )



Irradiance or flux density at the image plane

$$I \propto \left(\frac{D}{f}\right)^2$$

where  $D/f$  is known as the **relative aperture**

Diameter  
focal length = relative aperture

# ***f*-Number**

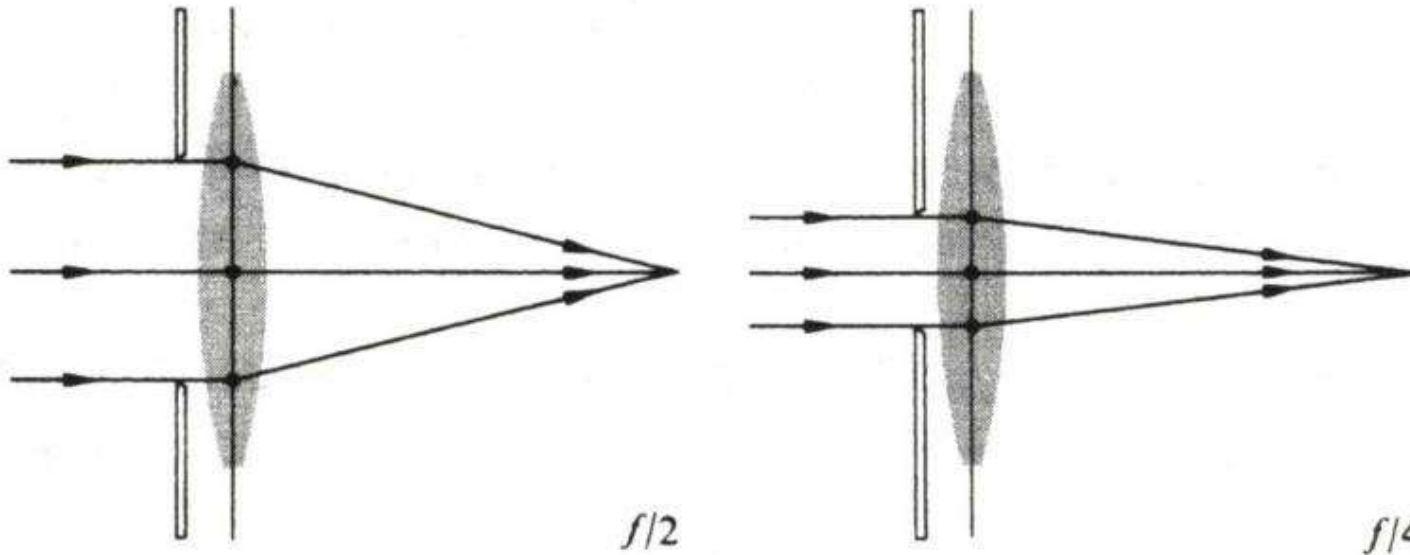
焦比

Focal ratio or f-number ( $f/\#$ ): inverse of relative aperture

$$f/\# = \frac{f}{D}$$

Example: a lens with a 25 mm aperture and a 50 mm focal length

$$f/\# = \frac{50}{25} = 2$$



A smaller f-number clearly permits more light to reach the image plane.

# Application

Camera lenses are usually specified by their focal lengths and the largest possible apertures



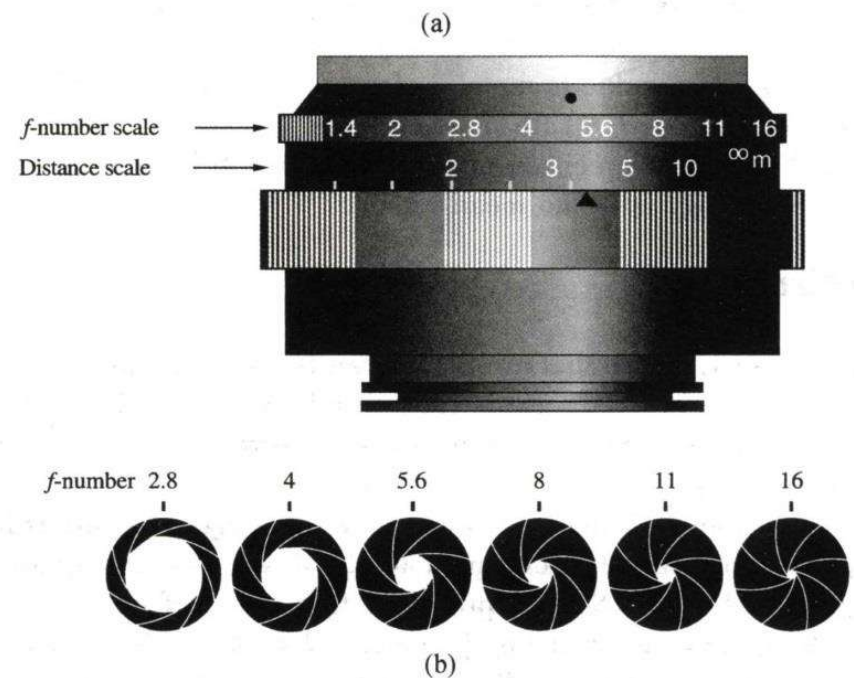
$$\text{Irradiance} \propto \left(\frac{D}{f}\right)^2 = \text{square of relative aperture}$$

$$\text{Exposure time} \propto \left(\frac{f}{D}\right)^2 = \text{square of } f/\#$$

The speed of the lens

Lens diaphragms have f-number markings of 1, 1.4, 2, 2.8, 4, 5.6, 8, 11, 16, 22, and so on.

The same amount of light will reach the film whether the camera is set for f/1.4 at 1/500<sup>th</sup> of a second, f/2 at 1/250<sup>th</sup> of a second, or f/2.8 at 1/125<sup>th</sup> of a second.



## 4.4 Mirrors

A mirror: a piece of black glass or a finely polished metal surface



Au mirror



Al mirror

For avoiding the losses due to metal surfaces, mirrors formed of multilayered dielectric films are used.

Broadband multilayer dielectric mirrors at 485-695 nm





# Planar Mirrors

Law of reflection  $\theta_i = \theta_r$

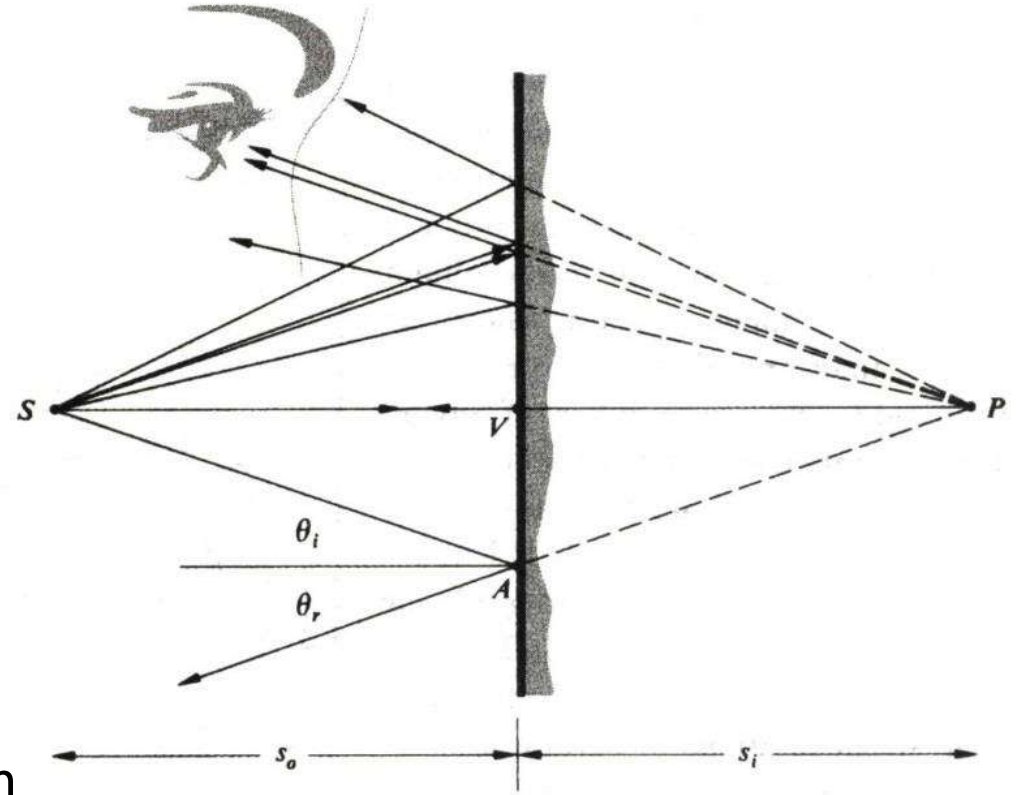
$$\theta_i = \theta_r$$

The image P and object S are equidistant from the surface

$$|s_o| = |s_i|$$

Follow the idea of virtual object and image distances being negative.

Definition:  $s_o$  and  $s_i$  are negative when they lie to the right of the vertex V.



Gaussian lens equation

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

Transverse magnification

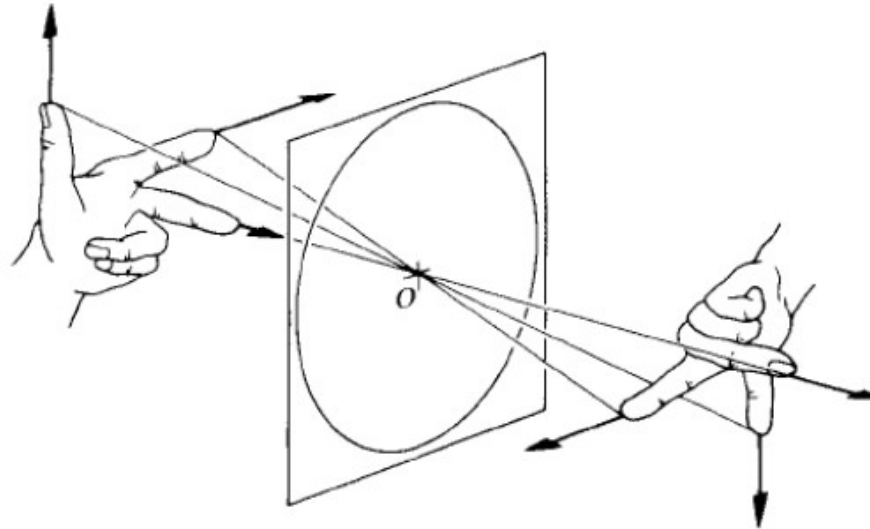
$$M_T = -\frac{s_i}{s_o} = +1$$

A life-size, virtual, erect image

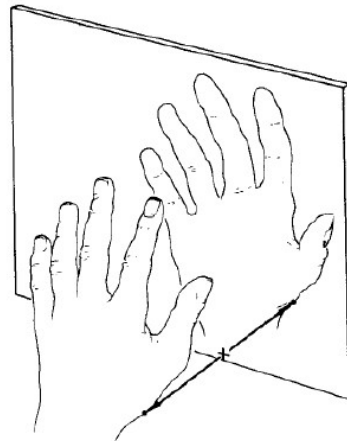


# Reversion (反转) and Inversion (反演)

Reversion:  $180^\circ$  rotation about the optical axis

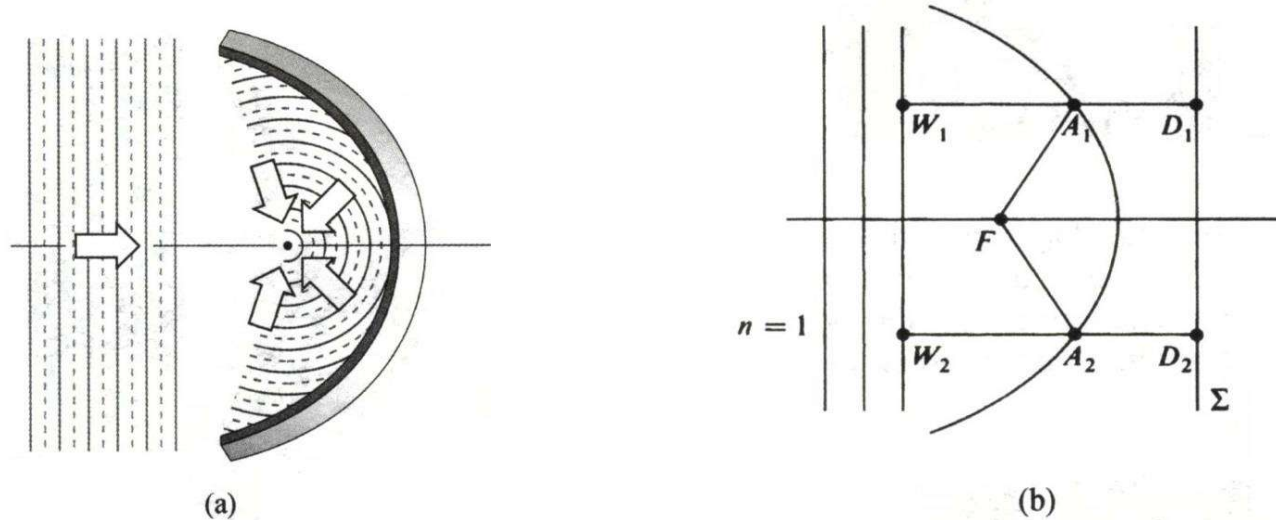


Inversion: convert a right-handed coordinate system in the object space into a left-handed one in the image space.



# Aspherical Mirrors

Let's determine the configuration a mirror must have if an incident plane wave is to be reformed upon reflection into a converging spherical wave.



The optical path lengths for all rays must be equal:

$$OPL = \overline{W_1 A_1} + \overline{A_1 F} = \overline{W_2 A_2} + \overline{A_2 F}$$

A plane  $\Sigma$  parallel to the incident wavefronts

$$\overline{W_1 A_1} + \overline{A_1 D_1} = \overline{W_2 A_2} + \overline{A_2 D_2}$$

There will be a surface for which  $\overline{A_1 F} = \overline{A_1 D_1}$  and  $\overline{A_2 F} = \overline{A_2 D_2}$ , it is a **parabola**.  
(抛物线)

# Application

Automobile headlight reflectors



Microwave horns



Communications antennas



# Spherical Mirrors

Equation for circular cross section of a sphere

$$y^2 + (x - R)^2 = R^2$$

Solution (concern the value  $x$  less than  $R$ )

$$x = R - (R^2 - y^2)^{1/2}$$

Expansion in binomial series

$$x = \frac{y^2}{2R} + \frac{1y^4}{2^2 2! R^3} + \frac{1 \cdot 3y^6}{2^3 3! R^5} + \dots$$

The first term: parabola

Other terms: deviation

For parabola

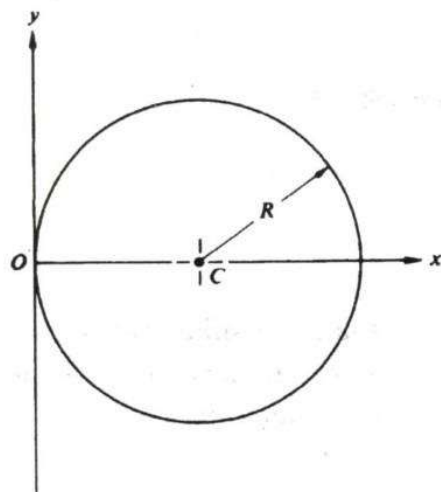
$$y^2 = 4fx$$

In paraxial region

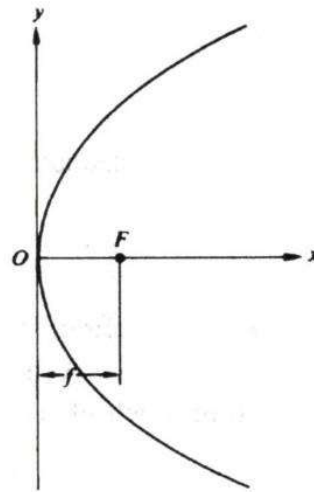
$$x = \frac{y^2}{2R}$$

where  $f = R/2$

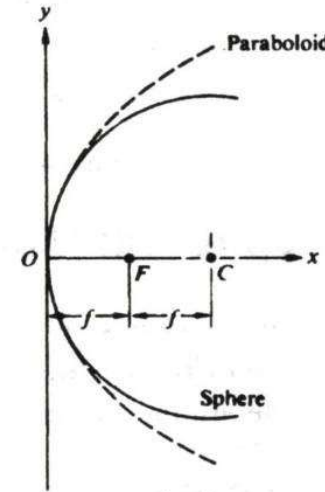
In the paraxial region, that is, in the immediate vicinity of the central axis, these two configurations will be essentially indistinguishable.



(a)



(b)



(c)

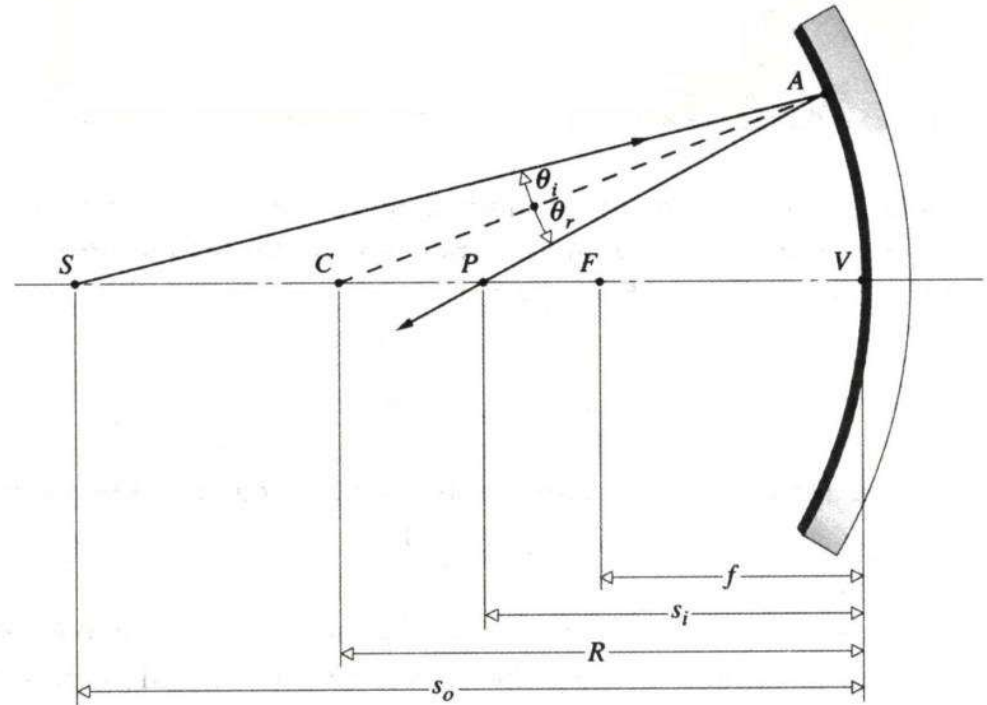
# The mirror formula

Since  $\theta_i = \theta_r$   $\frac{\overline{SC}}{\overline{SA}} = \frac{\overline{CP}}{\overline{PA}}$

Consider the sign of each item

$$\overline{SC} = s_o - |R| = s_o + R$$

$$\overline{CP} = |R| - s_i = -(s_i + R)$$



In the paraxial region

$$\overline{SA} \approx s_o$$

$$\overline{PA} \approx s_i$$

We have

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

## Sign Convention for Spherical Mirrors

| Quantity | Sign                     |                             |
|----------|--------------------------|-----------------------------|
|          | +                        | -                           |
| $s_o$    | Left of V, real object   | Right of V, virtual object  |
| $s_i$    | Left of V, real image    | Right of V, virtual image   |
| $f$      | Concave mirror           | Convex mirror               |
| $R$      | C right of V, convex     | C left of V, concave        |
| $y_o$    | Above axis, erect object | Below axis, inverted object |
| $y_i$    | Above axis, erect image  | Below axis, inverted image  |

## Focus

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

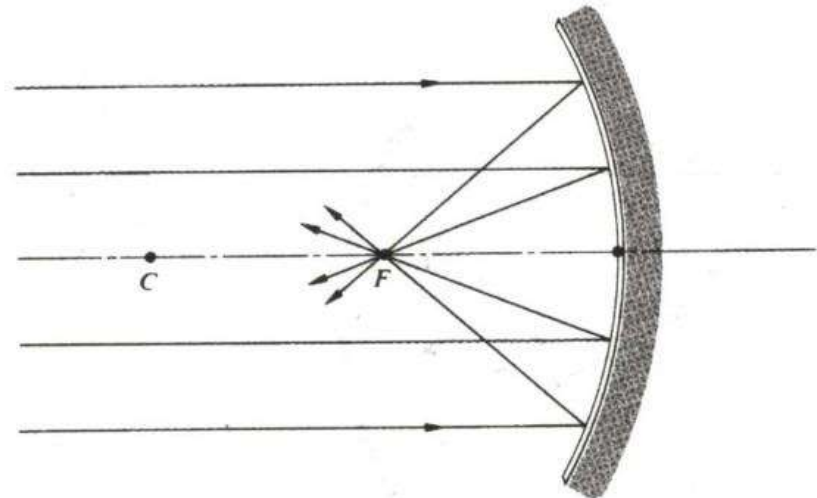
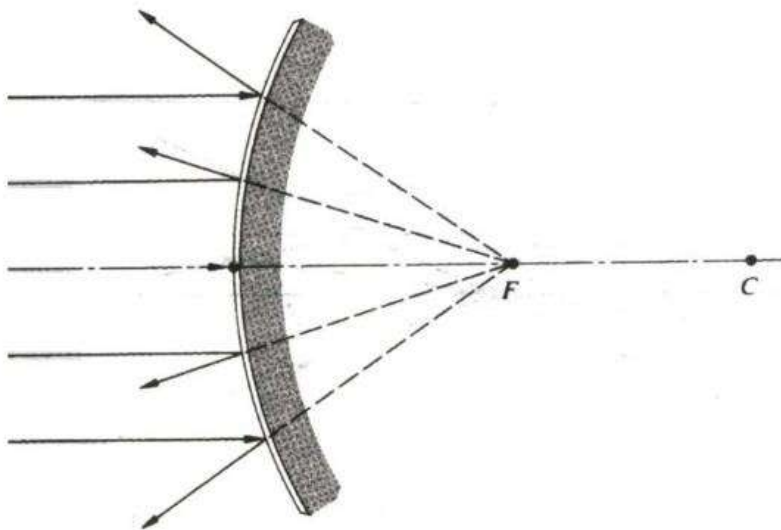
Primary or object focus

$$\lim_{s_i \rightarrow \infty} s_o = f_o = -\frac{R}{2}$$

Secondary or image focus

$$\lim_{s_o \rightarrow \infty} s_i = f_i = -\frac{R}{2}$$

Focus will be positive for concave mirrors ( $R < 0$ ) and negative for convex mirrors ( $R > 0$ )



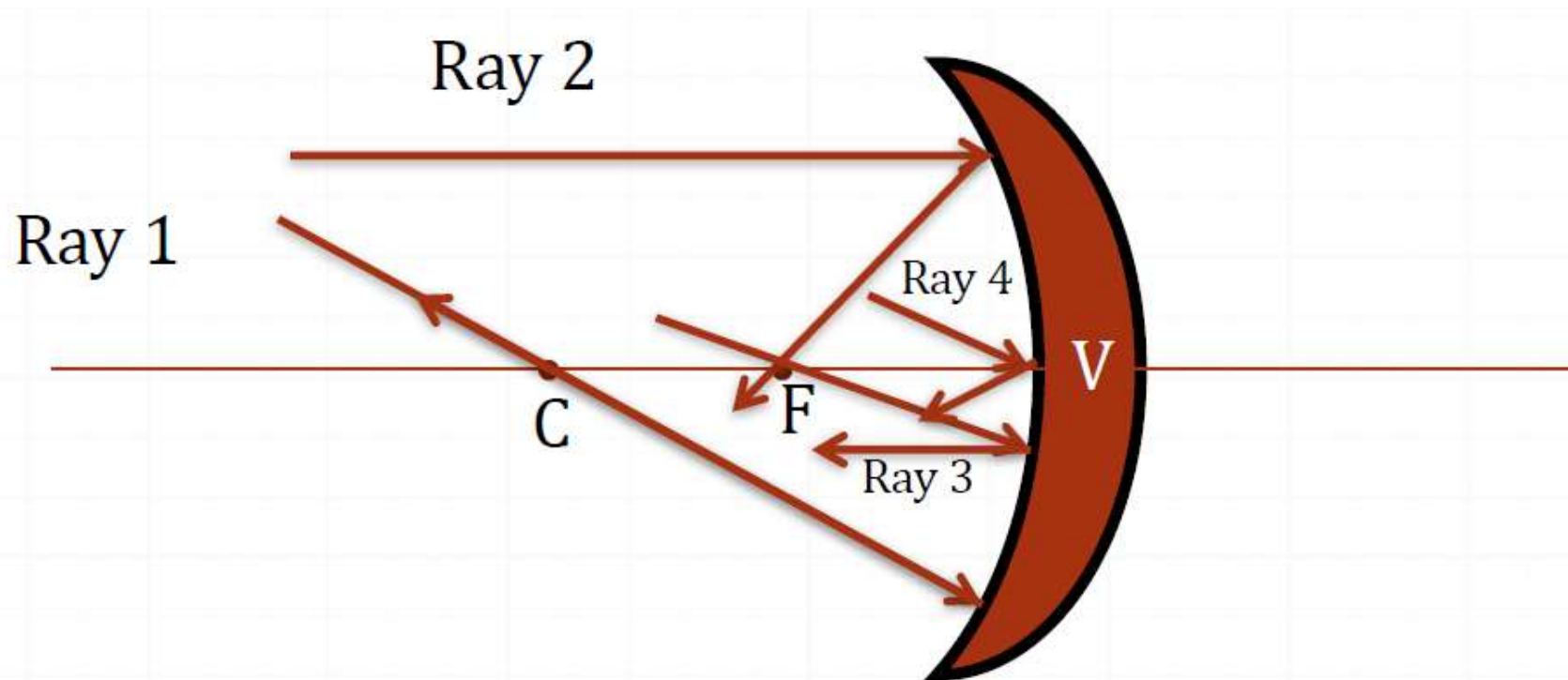
# Finite Imagery

Ray 1 passes through the center

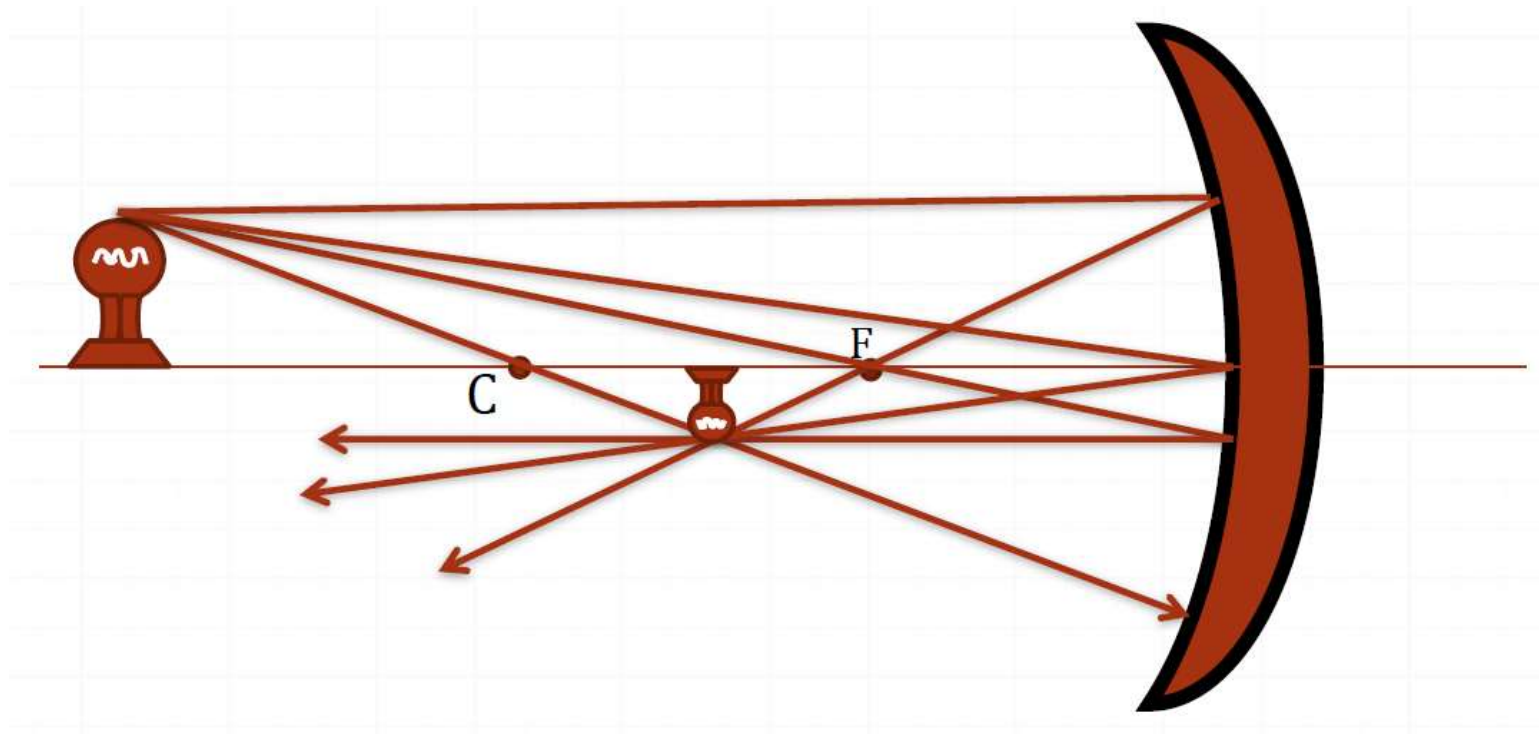
Ray 2 is parallel to the axis

Ray 3 passes through the focus

Ray 4 passes through the vertex



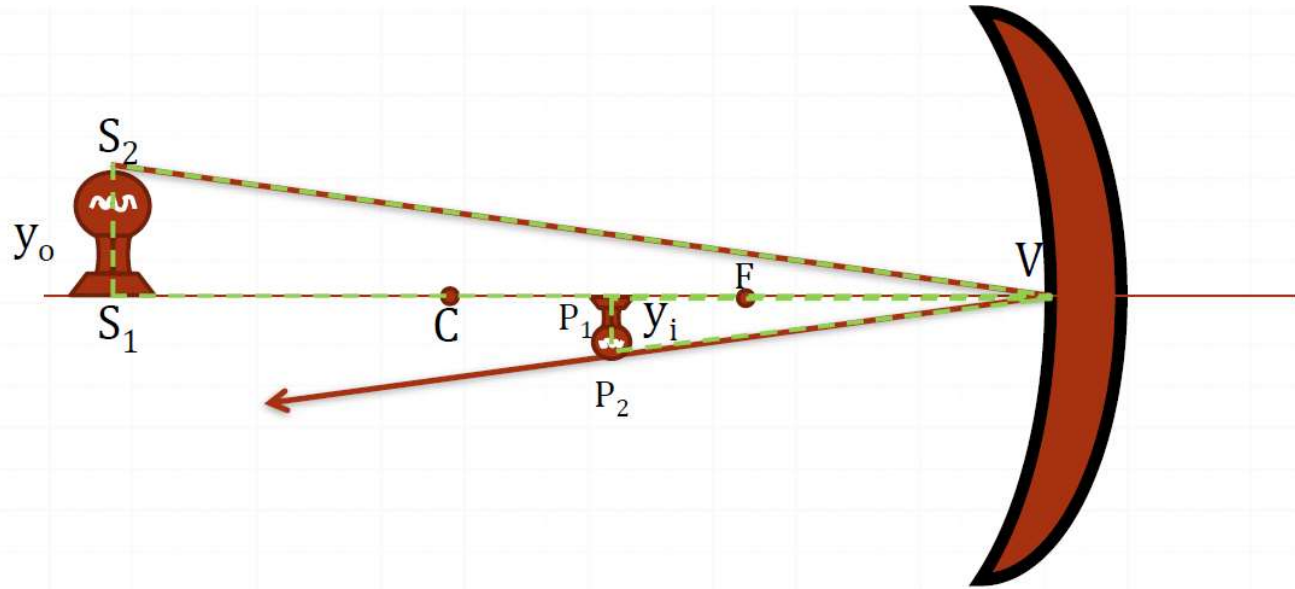
Within the restrictions of paraxial theory, any parallel off-axis bundle of rays will be focused to a point on the focal plane passing through F normal to the optical axis.





# Transverse magnification

$$M_T \equiv \frac{y_i}{y_o} = -\frac{s_i}{s_o}$$



The only equation that contains information about the structure of the optical element is that for  $f$ , it differs for the thin lens and spherical mirror.

The other functional expressions that relate  $s_o$ ,  $s_i$ , and  $f$  or  $y_o$ ,  $y_i$ , and  $M_T$  are precisely the same.

The only alteration in the previous sign convention is that  $s_i$  on the left of  $V$  is now taken as positive.

# The striking similarity between

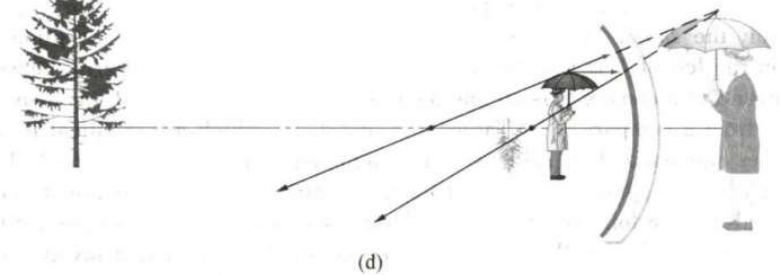
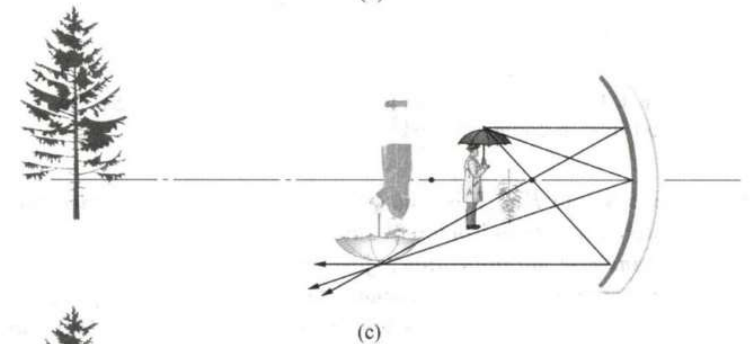
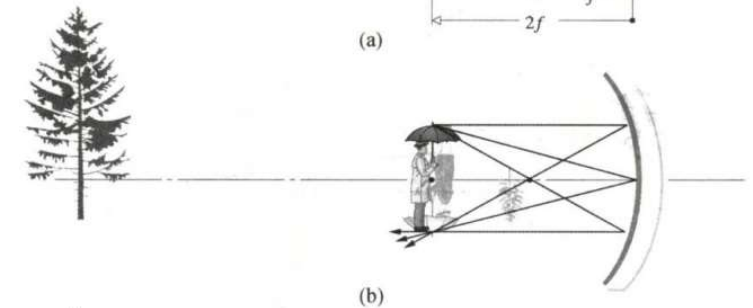
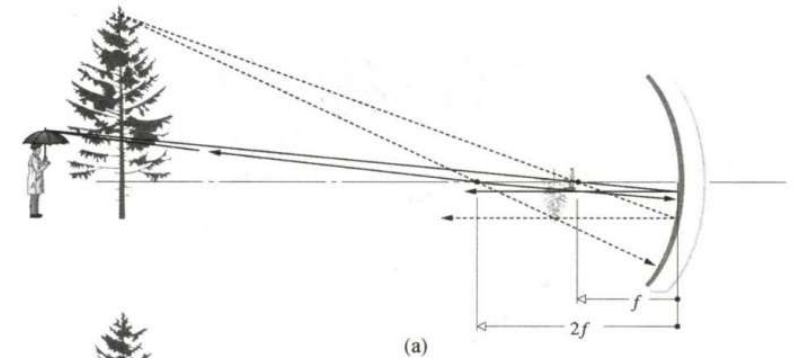
A concave mirror and a convex lens

A convex mirror and a concave lens

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

**Table 4.5 Images of Real Objects Formed by Spherical Mirrors**

| Concave             |         |                                  |             |               |
|---------------------|---------|----------------------------------|-------------|---------------|
| Object              |         | Image                            |             |               |
| Location            | Type    | Location                         | Orientation | Relative Size |
| $\infty > s_o > 2f$ | Real    | $f < s_i < 2f$                   | Inverted    | Minified      |
| $s_o = 2f$          | Real    | $s_i = 2f$                       | Inverted    | Same size     |
| $f < s_o < 2f$      | Real    | $\infty > s_i > 2f$              | Inverted    | Magnified     |
| $s_o = f$           |         | $\pm \infty$                     |             |               |
| $s_o < f$           | Virtual | $ s_i  > s_o$                    | Erect       | Magnified     |
| Convex              |         |                                  |             |               |
| Object              |         | Image                            |             |               |
| Location            | Type    | Location                         | Orientation | Relative Size |
| Anywhere            | Virtual | $ s_i  <  f $ ,<br>$s_o >  s_i $ | Erect       | Minified      |

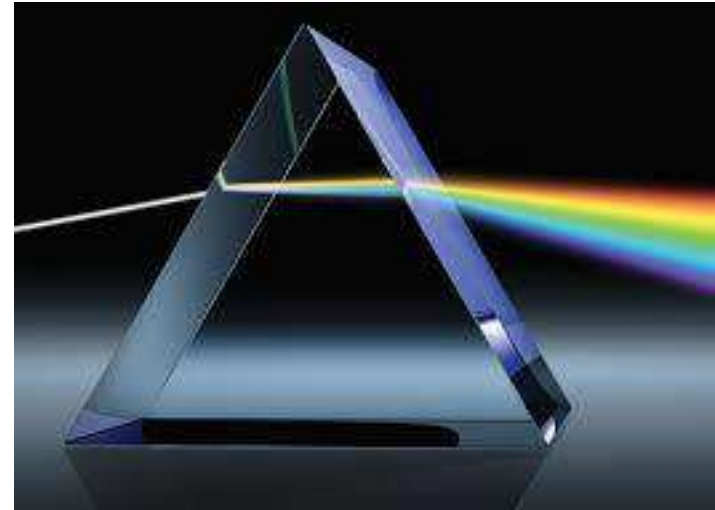


# Homework

□ 5.44, 5.46, 5.47, 5.52, 5.55, 5.60

# 4.5 Prisms

- ❑ Dispersive prisms
  - ✓ Separate the constituent frequency components in a polychromatic light beam.
  
- ❑ Reflecting prisms
  - ✓ Change the orientation of an image or direction of propagation of a beam



# Dispersing prisms

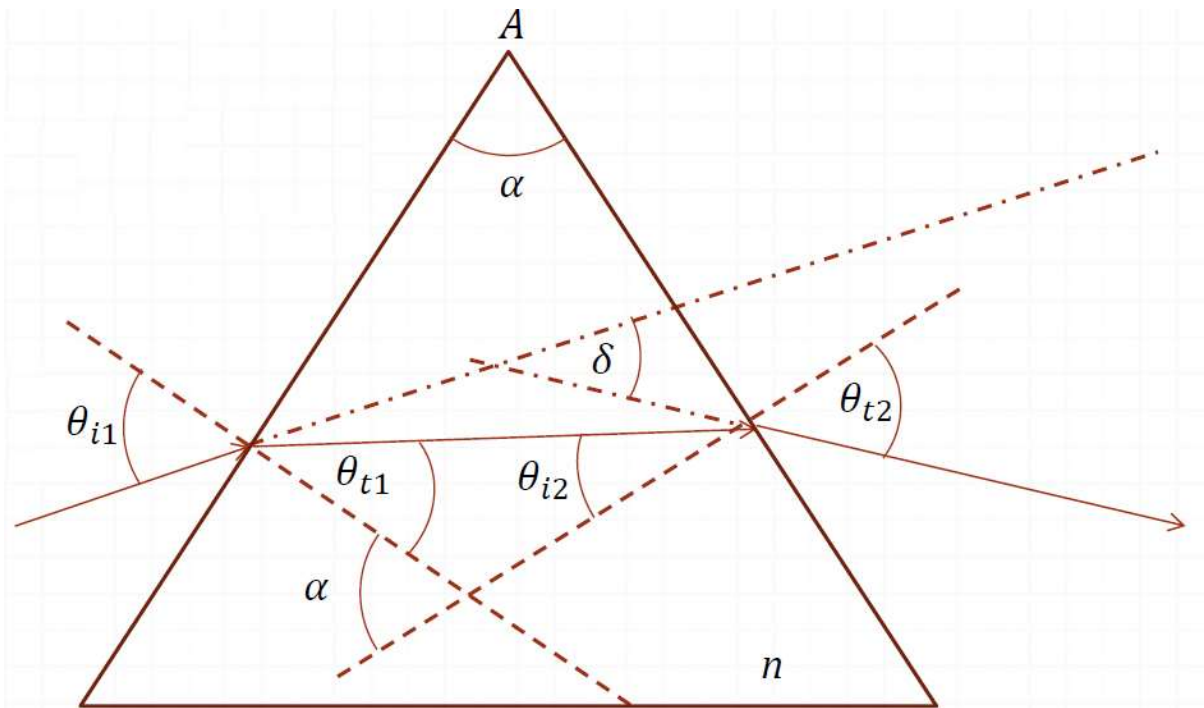
Angular deviation ( $\delta$ ): a ray entering a dispersing prism will emerge having been deflected from its original direction by an angle.

$$\delta = (\theta_{i1} - \theta_{t1}) + (\theta_{t2} - \theta_{i2})$$

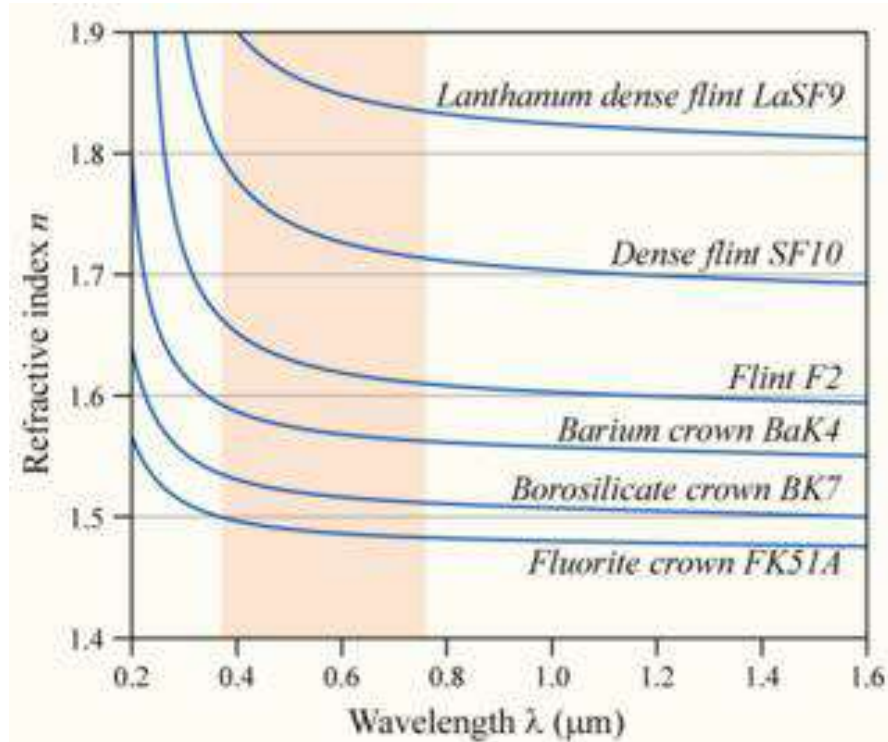
$$\text{Apex angle } \alpha = \theta_{t1} + \theta_{i2} \qquad \delta = \theta_{i1} + \theta_{t2} - \alpha$$

$$\delta = \theta_{i1} + \sin^{-1} \left[ \sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$

$\delta$  increases with  $n$ , which is itself a function of frequency, so we designate the deviation as  $\delta(\nu)$  or  $\delta(\lambda)$



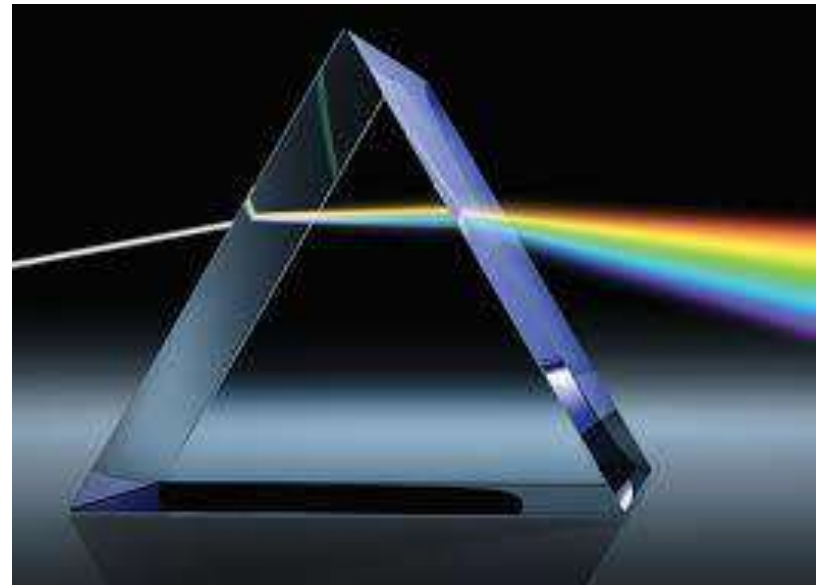
The wavelength dependence of the index of refraction for various materials



$$\delta = \theta_{i1} + \sin^{-1} \left[ \sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$

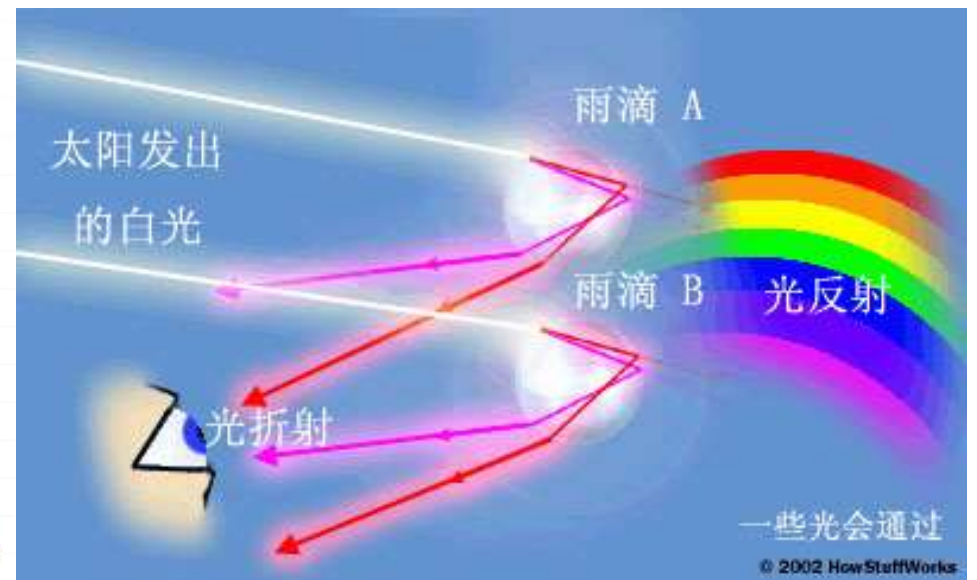
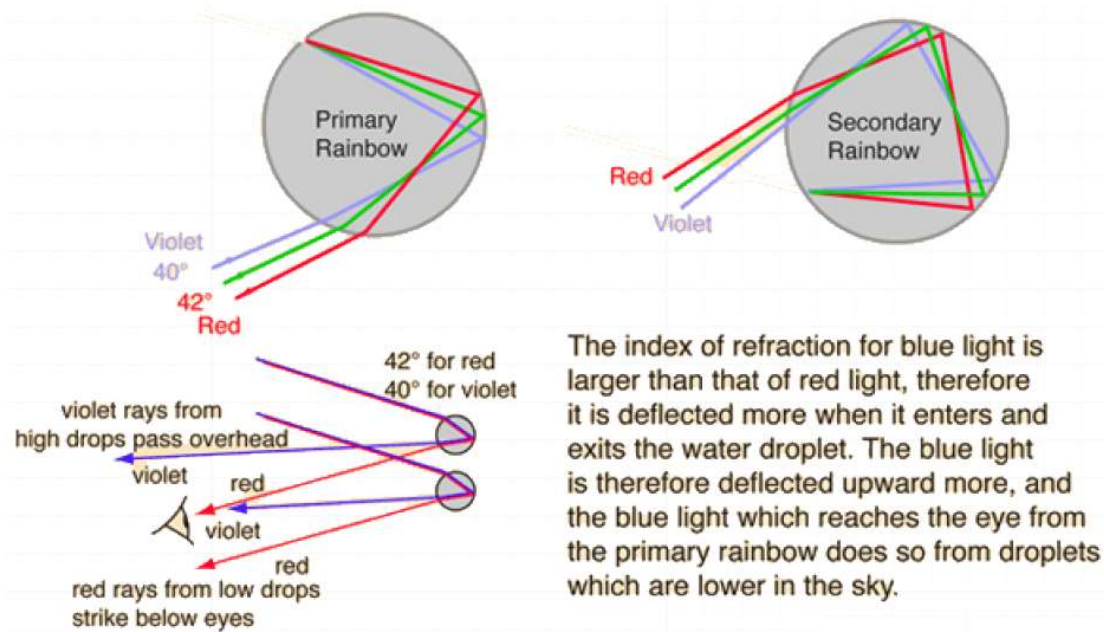


So  $\delta(\lambda)$  will be less for red light than it is for blue



**For most transparent dielectric of practical concern,  $n(\lambda)$  decreases as the wavelength increases across the visible.**





### Double rainbow with an Alexander's dark band

Storms paused just long enough for a spectacular display of light in Tucson—a glorious double rainbow with a highly visible Alexander's dark band.

—Gregory Quarles, OSA Fellow

30 Mar 2020

$$\delta = \theta_{i1} + \sin^{-1} \left[ \sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$

The deviation suffered by a monochromatic beam on traversing a given prism is a function only of the incident angle at the first face.

The smallest value of  $\delta$  is known as the **minimum deviation**

Starting from

$$\delta = \theta_{i1} + \theta_{t2} - \alpha$$

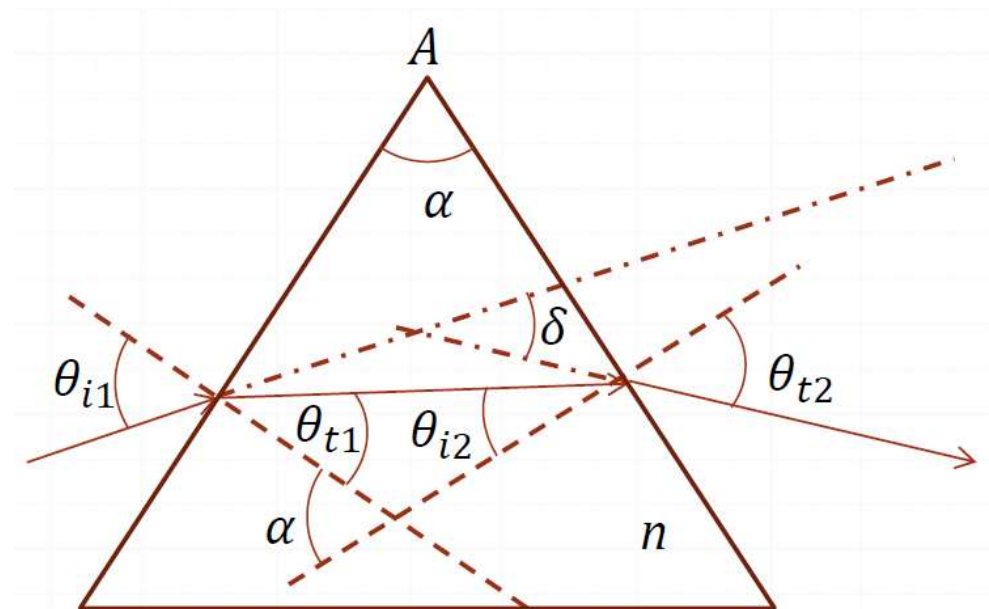
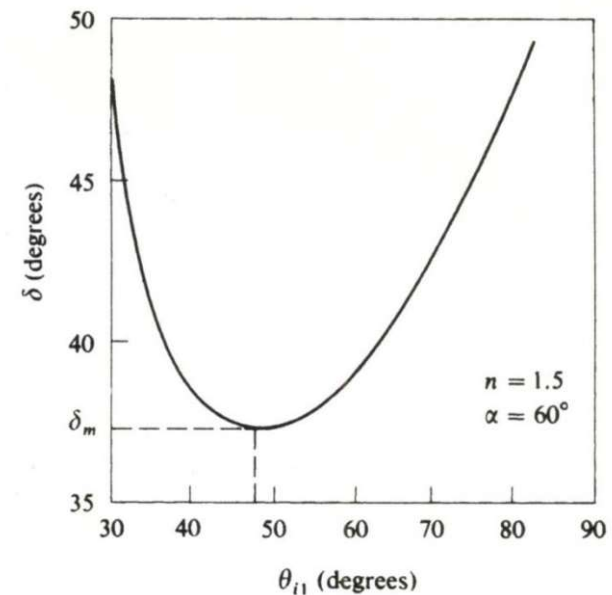
$$\alpha = \theta_{t1} + \theta_{i2}$$

Solution:

$$\theta_{i1} = \theta_{t2}$$

$$\theta_{t1} = \theta_{i2}$$

The ray for which the deviation is a minimum traverses the prism symmetrically, that is parallel to its base.





In the case when  $\delta = \delta_m$

$$\delta_m = \theta_{i1} + \theta_{t2} - \alpha$$

$$\theta_{i1} = \theta_{t2}$$

$$\theta_{i1} = (\delta_m + \alpha)/2$$

$$\alpha = \theta_{t1} + \theta_{i2}$$

$$\theta_{t1} = \theta_{i2}$$

$$\theta_{t1} = \alpha/2$$

Snell's law

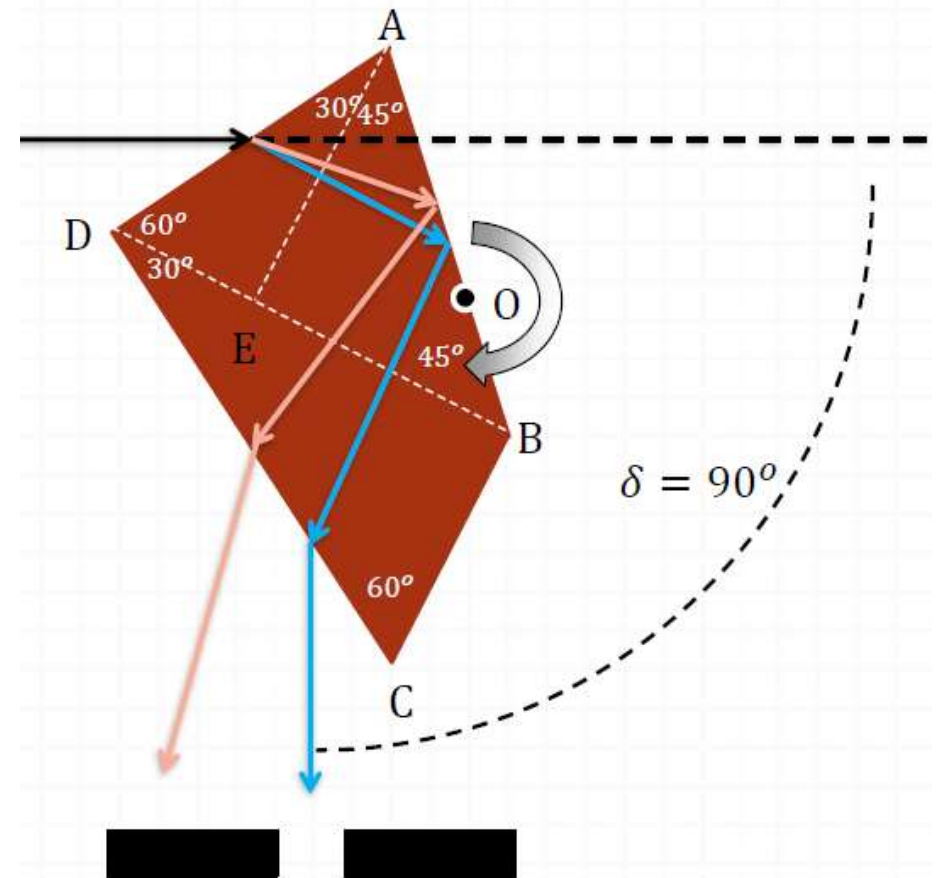
$$n = \frac{\sin \theta_{i1}}{\sin \theta_{t1}} = \frac{\sin (\delta_m + \alpha)/2}{\sin \alpha/2}$$

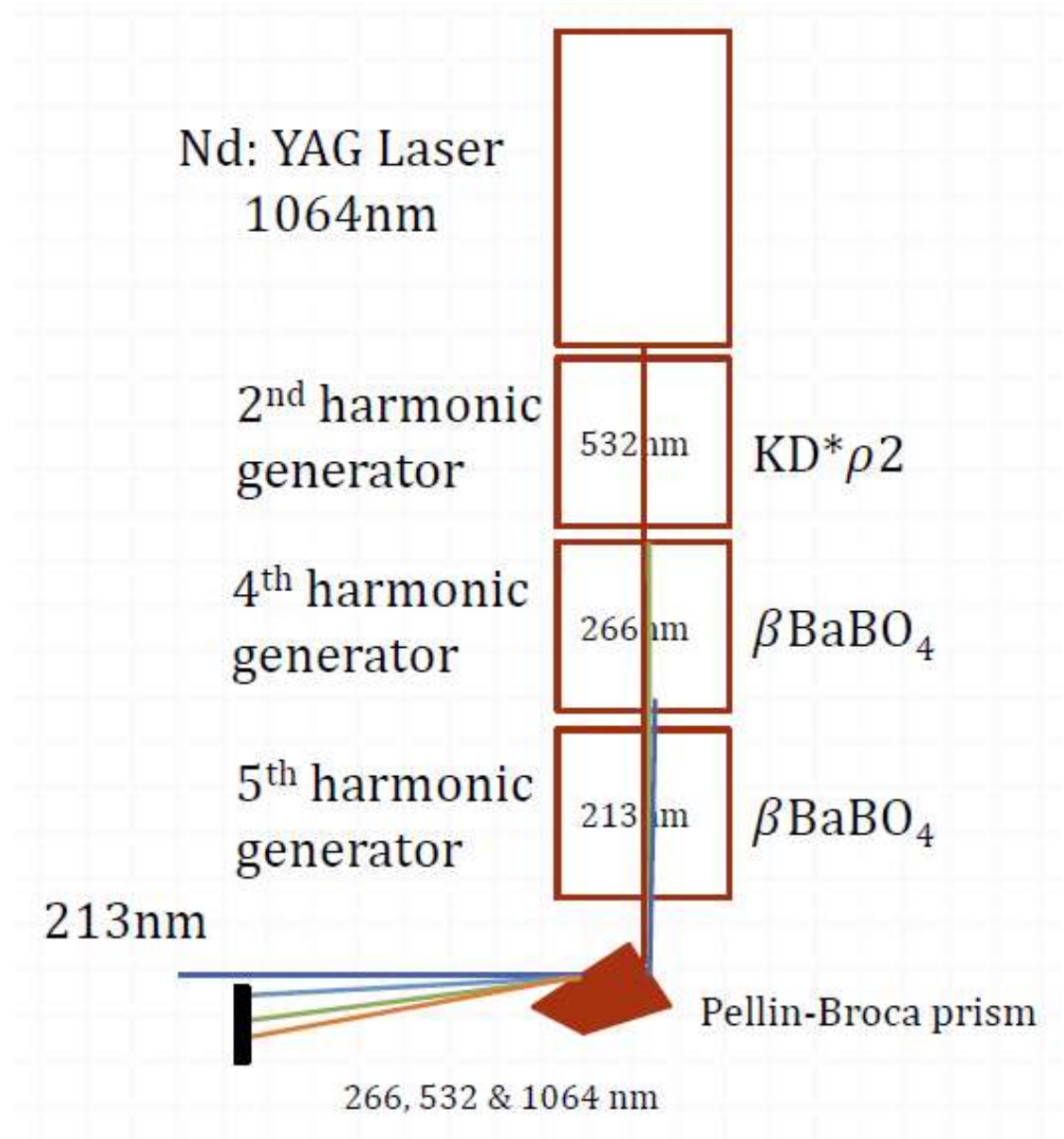
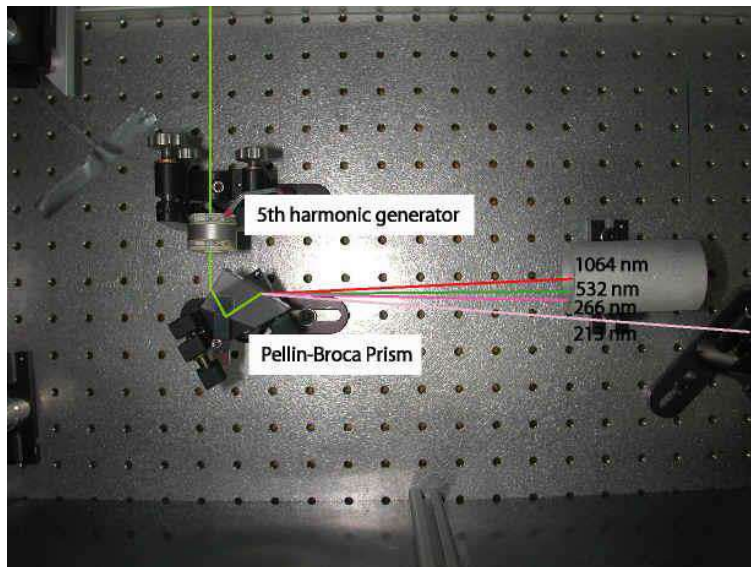
This equation forms the basis of technique for determining the refractive index.

# Constant-deviation dispersing prisms (恒偏向色散棱镜)

## Pellin-Broca (佩林-布洛卡) prism

- Suppose that in the position shown a single monochromatic ray of wavelength  $\lambda$  traverses the component prism DAE symmetrically, thereafter to be reflected at  $45^\circ$  from face AB. The ray will then traverse prism CDB symmetrically, having experienced a total deviation of  $90^\circ$ .
- As the prism is rotated around a point O, the selected wavelength which is deviated by  $90^\circ$  is changed without changing the geometry of relative positions of the input and output beams.





# Reflecting prisms

Reflecting prisms: dispersion is not desirable, which means that deviation is independent of  $\lambda$ .

The angle of deviation ( $\delta$ ):

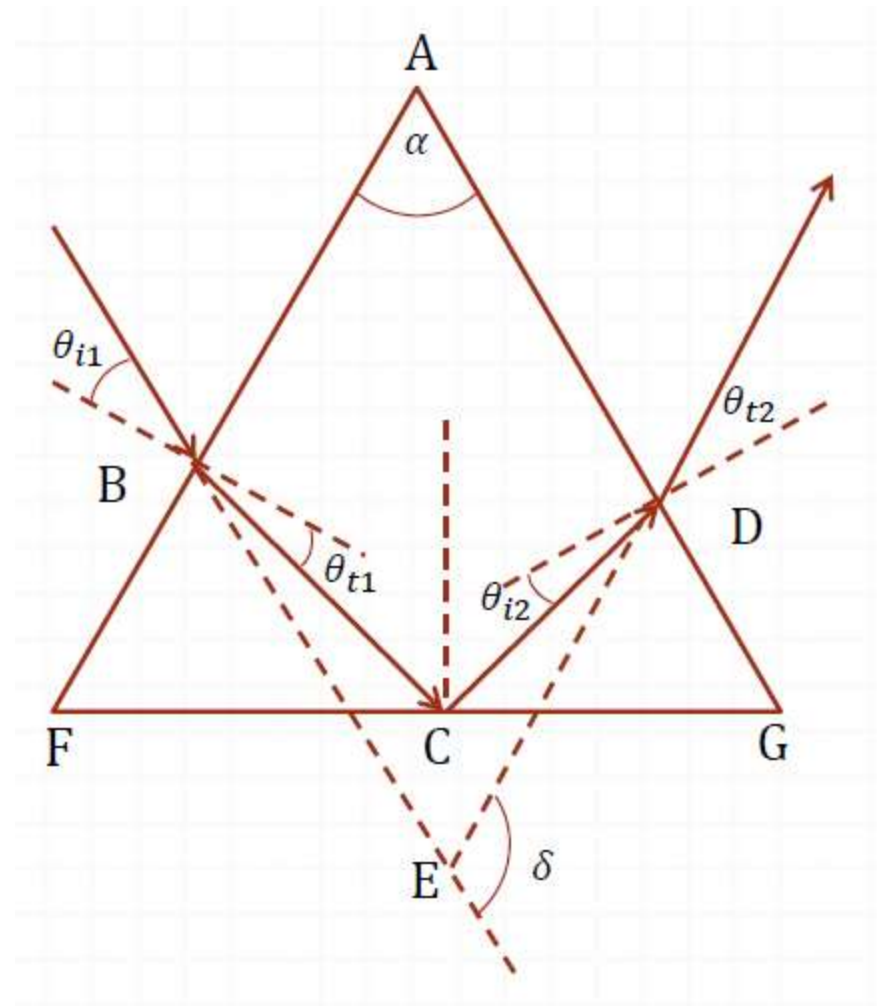
$$\delta = \theta_{i1} + \theta_{t2} + \alpha$$

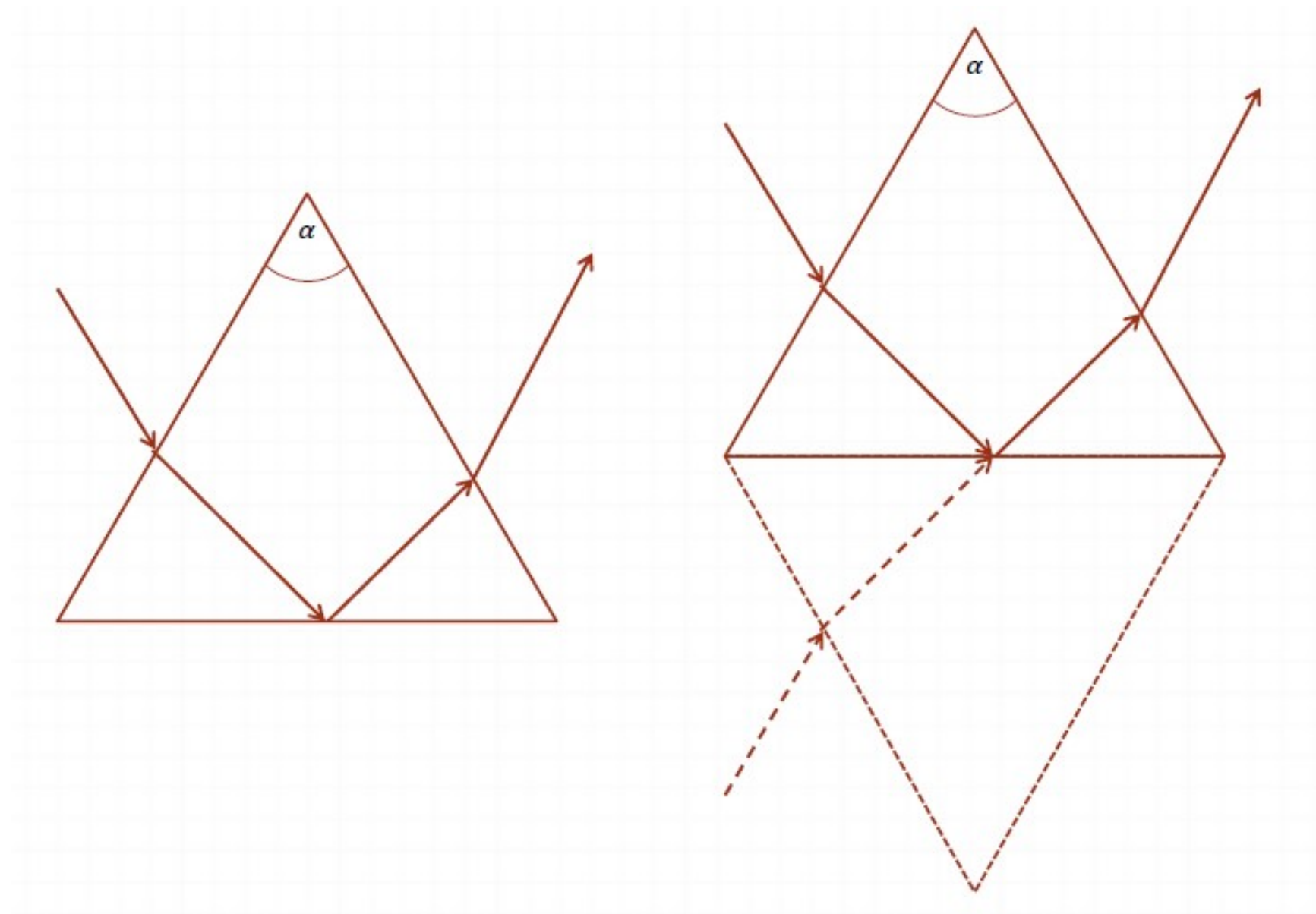
$$\delta = 2\theta_{i1} + \alpha$$

which is independent of both  $\lambda$  and  $n$ .

That is to say, it is **achromatic**.

Purpose: change the direction of propagation or the orientation of the image, or both.





The image of the incident ray emerges parallel to itself, regardless of wavelength.



# Reflecting prisms

## Image Inversion and Reversion by Right-Angle Prisms

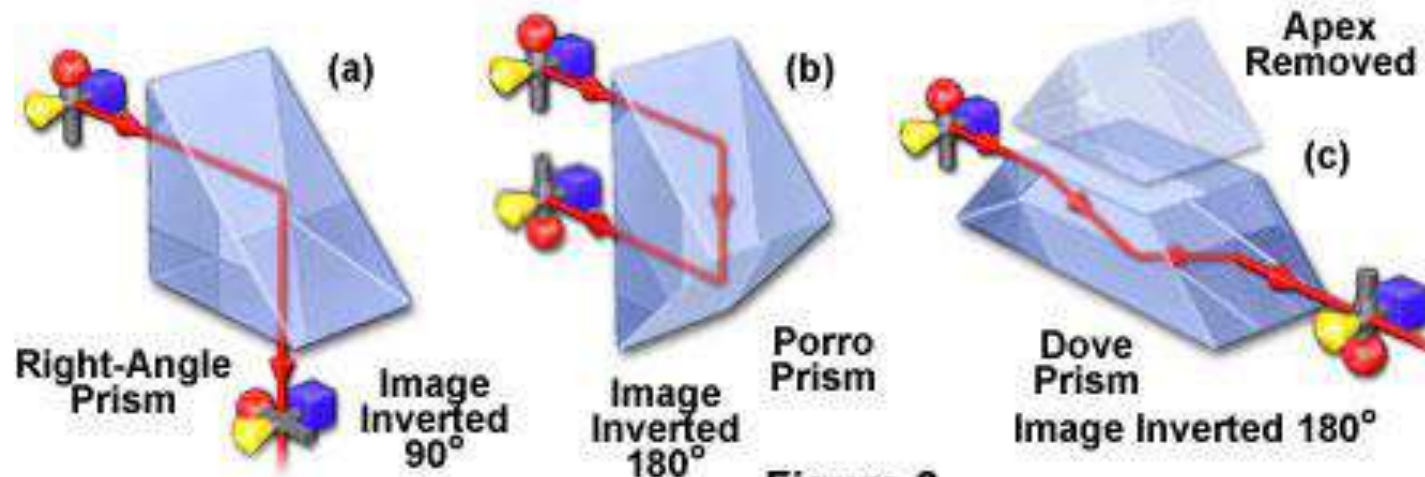


Figure 2

## Image Rotation, Deviation, and Displacement by Common Prisms

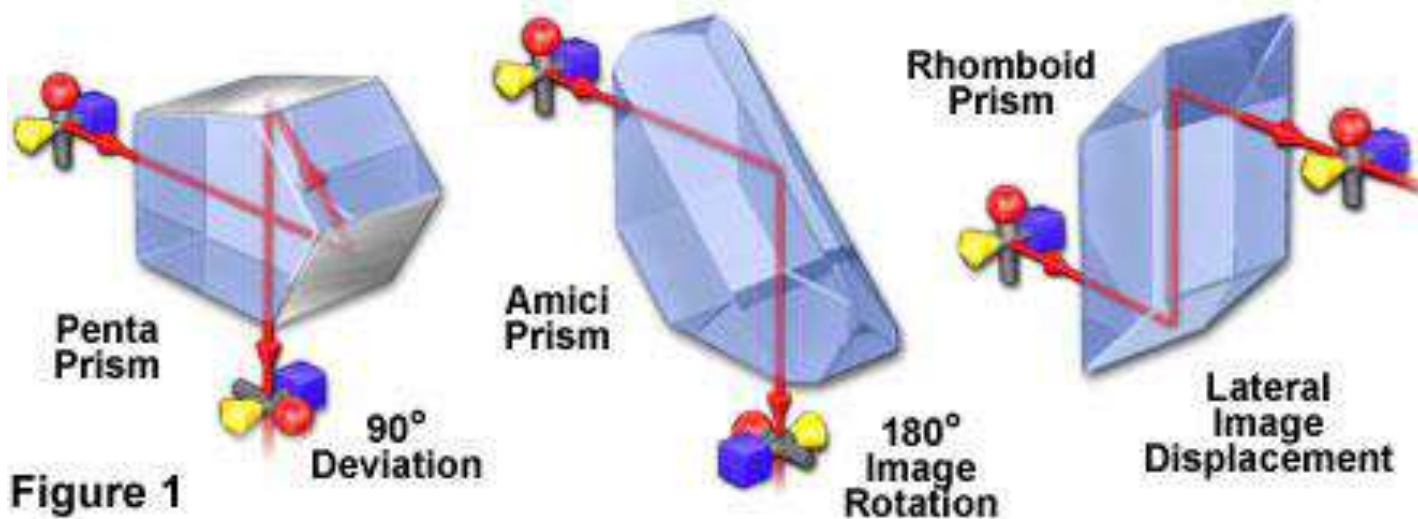


Figure 1

## 4.6 Fiber optics

Communication method:

Electric current, microwave, **light**.

**Fiber:** because of its low-loss transmission, high-information-carrying capacity, small size and weight, immunity to electromagnetic interference, unparalleled signal security, and the abundant availability of the required raw materials (i.e., ordinary sand), ultra pure glass fibers have become the premier communications medium.

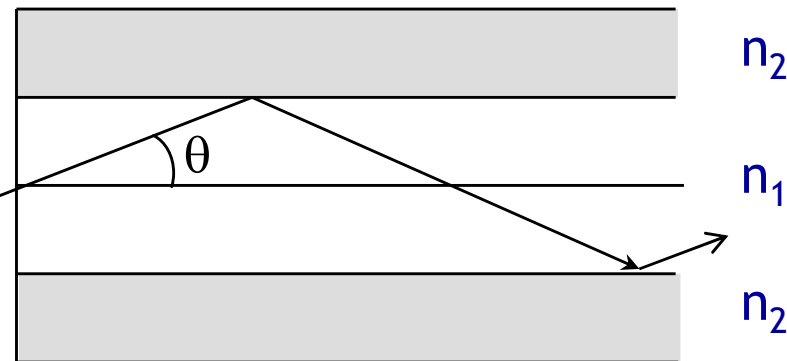
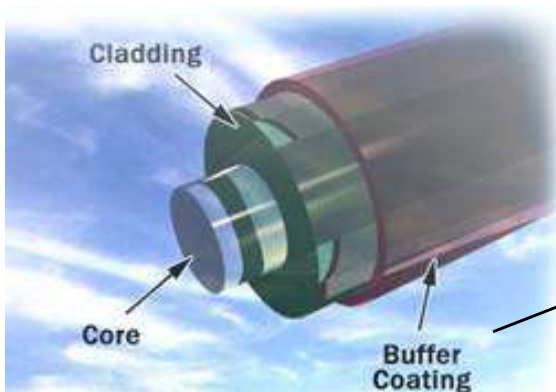


# 1966 Breakthrough of fiber optics

**Charles K. Kao**, working with George Hockham at Standard Telecommunication Laboratories in Harlow, England, makes a discovery that leads to a breakthrough in fiber optics. He calculates how to transmit light over long distances via optical glass fibers, deciding that, with a fiber of purest glass, it would be possible to transmit light signals over a distance of 100 km, compared with only 20 m for the fibers available in the 1960s. Kao receives a 2009 Nobel Prize in physics for his work.



1933—2018



The path length:

$$\ell = L / \cos \theta_t$$

From Snell's law:

$$\ell = n_f L (n_f^2 - \sin^2 \theta_i)^{-1/2}$$

The number of reflections N:

$$N = \frac{\ell}{D / \sin \theta_t} \pm 1 = \frac{L \sin \theta_i}{D (n_f^2 - \sin^2 \theta_i)^{1/2}} \pm 1$$

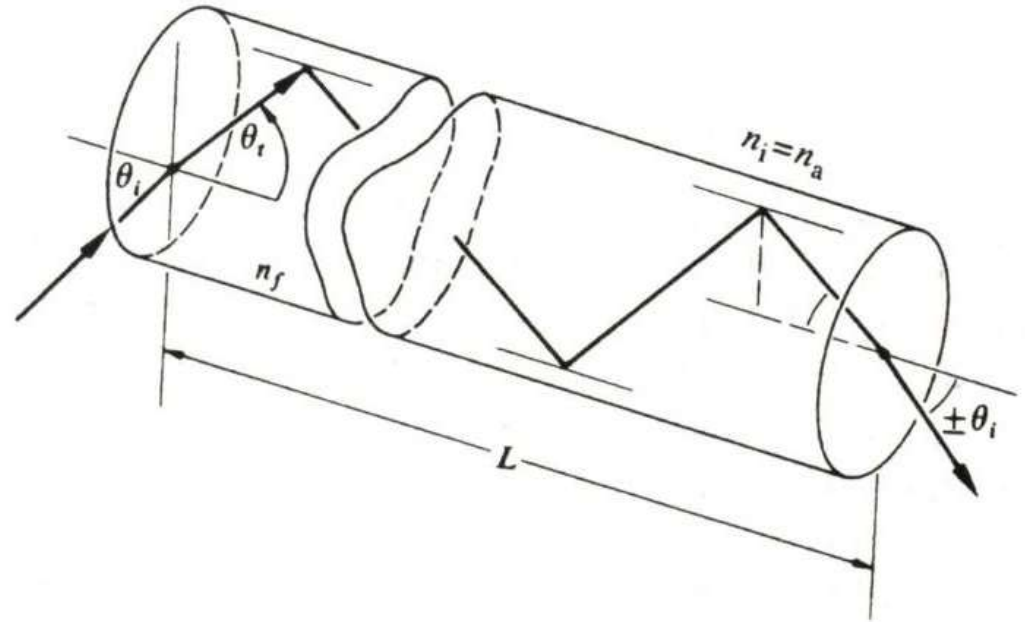
Example:  $D = 50 \mu\text{m}$ ;  $n_f = 1.6$ ;  $\theta_i = 30^\circ$   
 $N/L = 2000$  reflection per foot.

Leakage of light:

If the surface of a single fiber is covered by moisture, dust, oil (via FTIR)

If large number of fibers are packed in close proximity (via FTIR), i.e., cross-talk

Solution: Each fiber is covered with a transparent sheath of lower index.



# Numerical aperture

A fiber core has an index  $n_f$ , and the cladding has an index  $n_c$ .

$$\sin \theta_c = \frac{n_c}{n_f} = \sin(90^\circ - \theta_t)$$

Thus

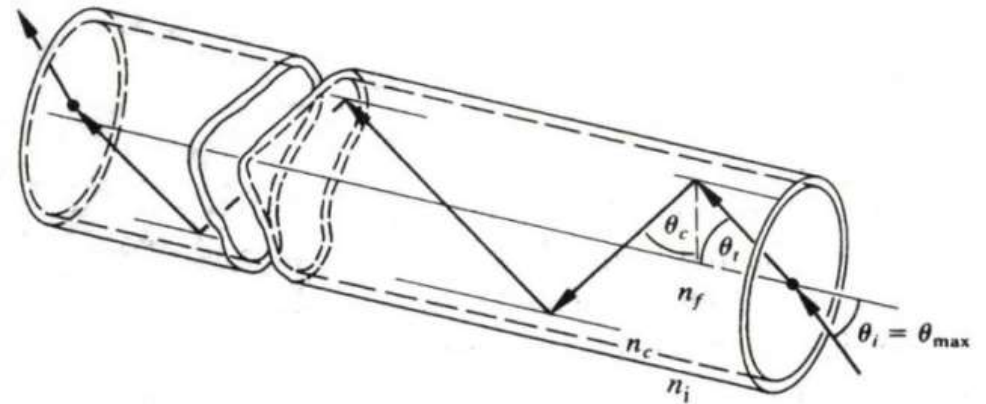
$$\frac{n_c}{n_f} = \cos \theta_t = (1 - \sin^2 \theta_t)^{1/2}$$

Solution:

$$\sin \theta_{max} = \frac{1}{n_i} (n_f^2 - n_c^2)^{1/2}$$

**Numerical aperture or NA:**  $n_i \sin \theta_{max}$

Fibers with a wide variety of numerical apertures, from about 0.2 up to and including 1, are commercially obtainable.



Bundles of free fibers whose ends are bound together, ground, and polished from flexible light guides. If no attempt is made to align the fibers in an ordered array, they form an ***incoherent bundle***.

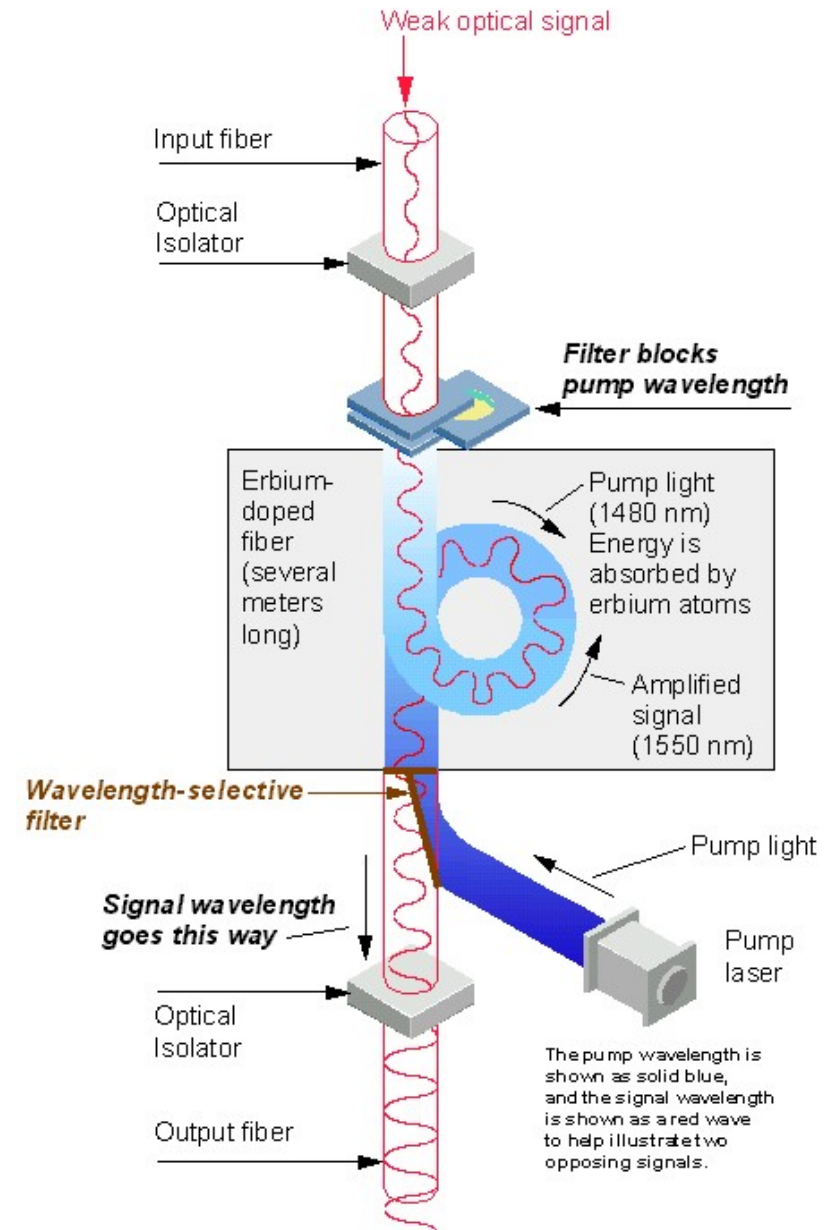
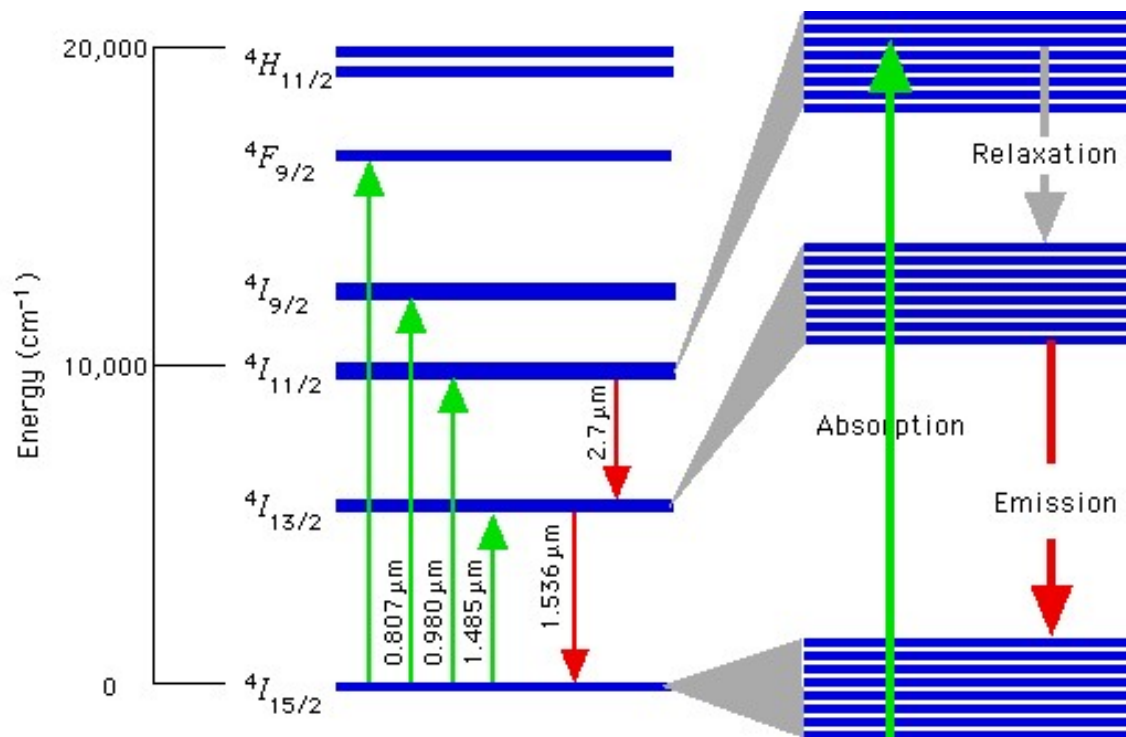


When the fibers are carefully arranged so that their terminations occupy the same relative positions in both of the bound ends of the bundle, it is said to be ***coherent***.



# Fiberoptic Communications Technology

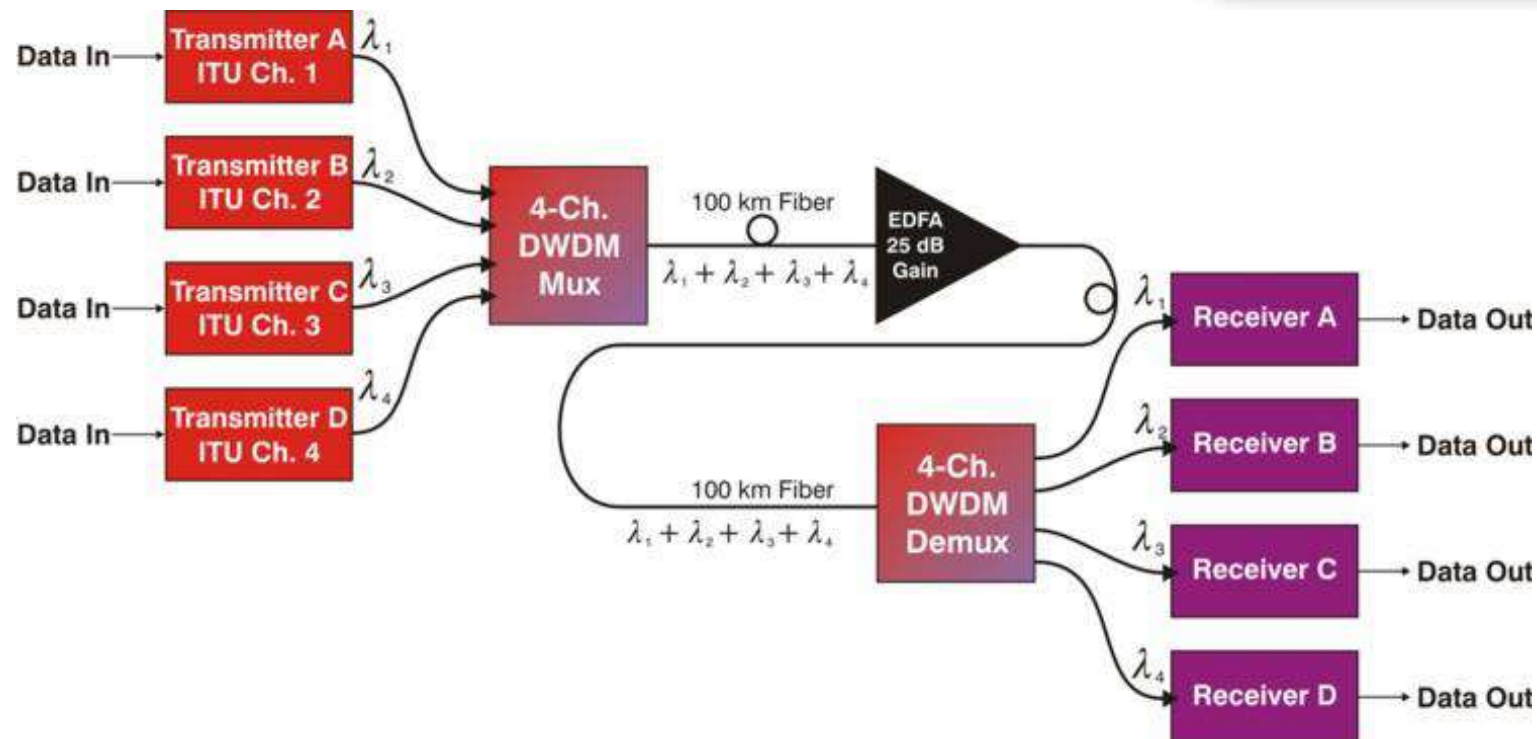
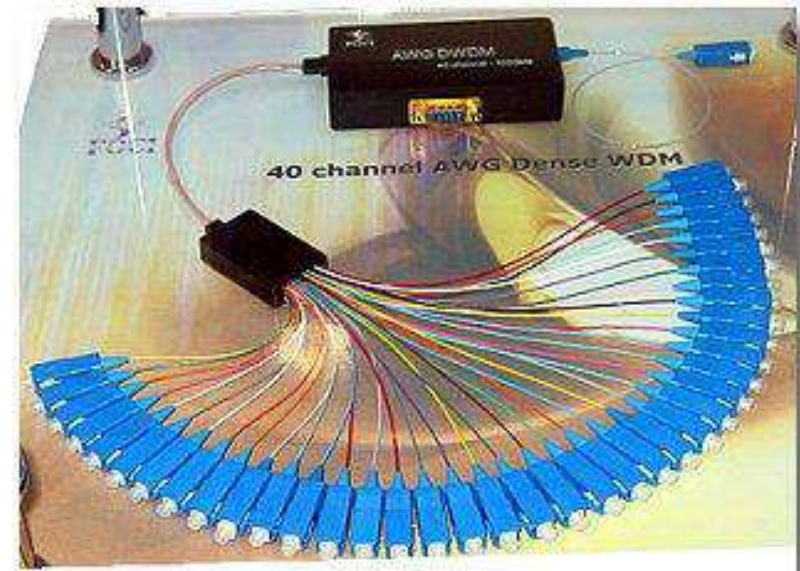
- ❑ **Attenuation:** due to impurities (especially the ions of iron, nickel and copper) and contamination of the fibers.
- ❑ **Innovational solution: erbium-doped fiber amplifiers**



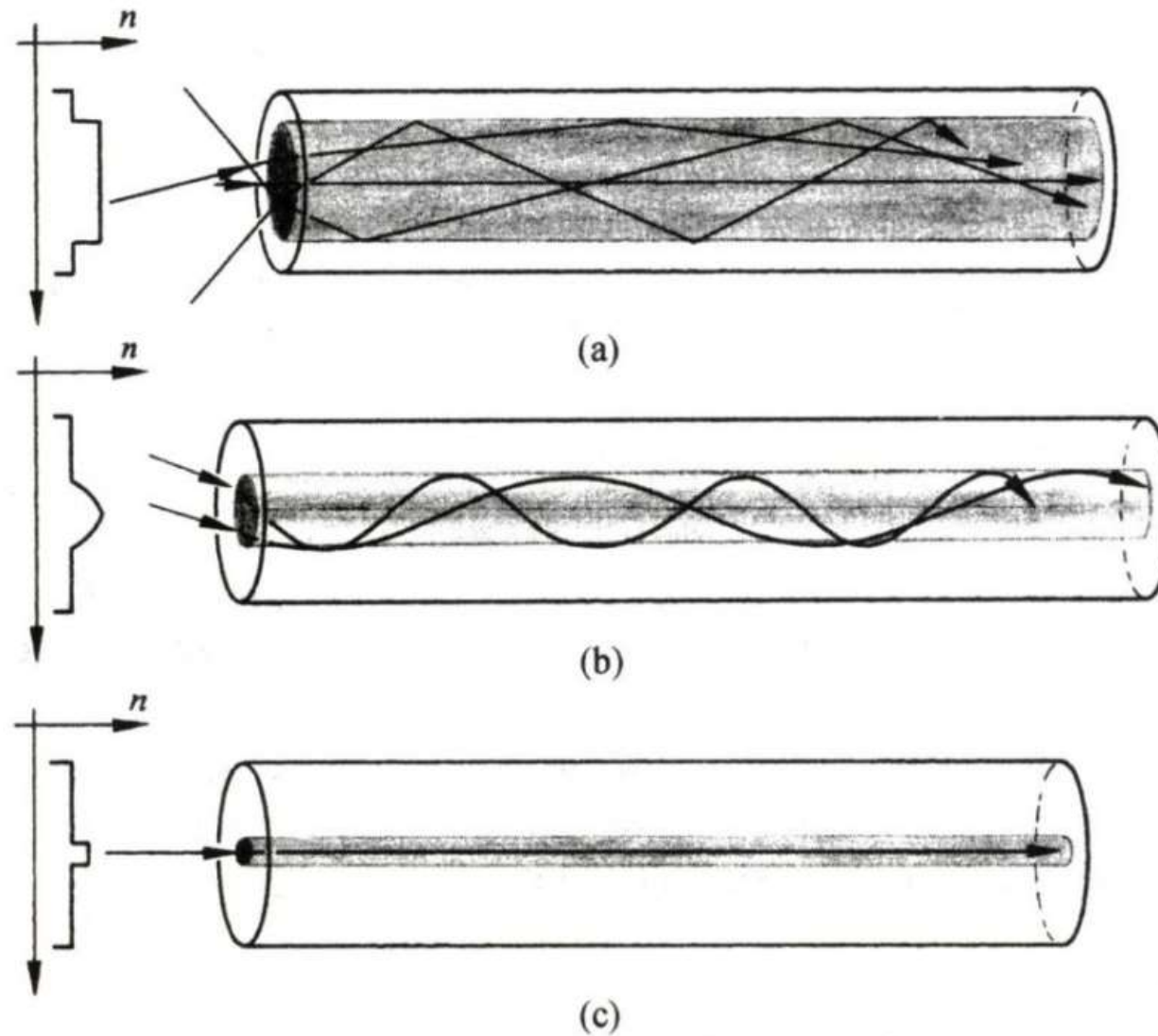


## The second innovational technique:

**Dense wavelength division multiplexing:** all the optical signals transmitted at the same time over the same fiber at different frequencies.



# Three major fiber configurations



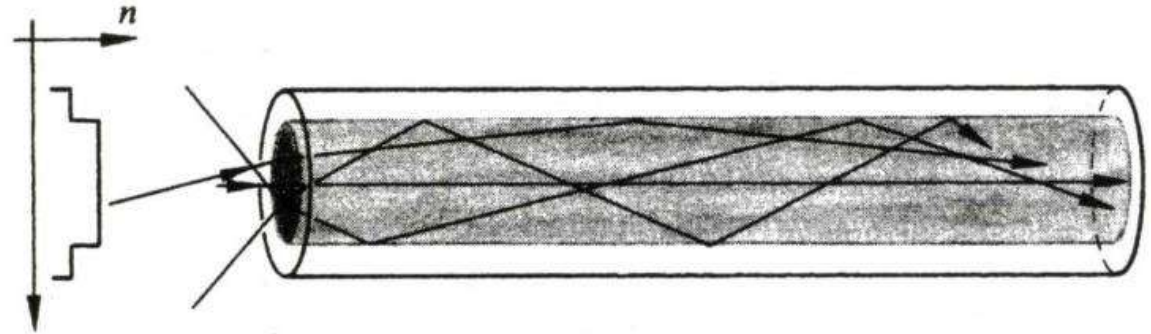
- (a) Multimode step-index fiber.
- (b) Multimode graded-index fiber.
- (c) Single-mode step-index fiber.



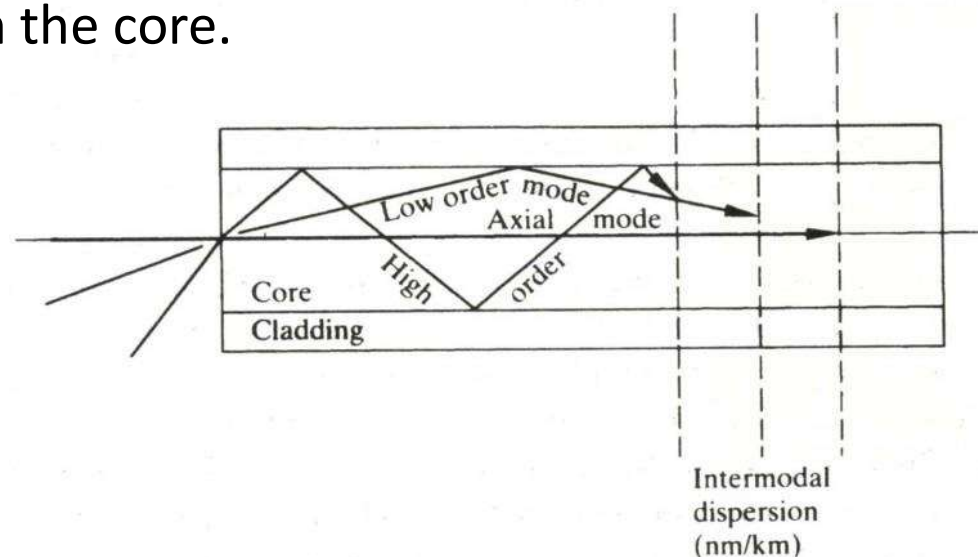
## Stepped-index fiber:

A homogeneous core of roughly 50 to 200  $\mu\text{m}$  and cladding typically 20  $\mu\text{m}$  thick.

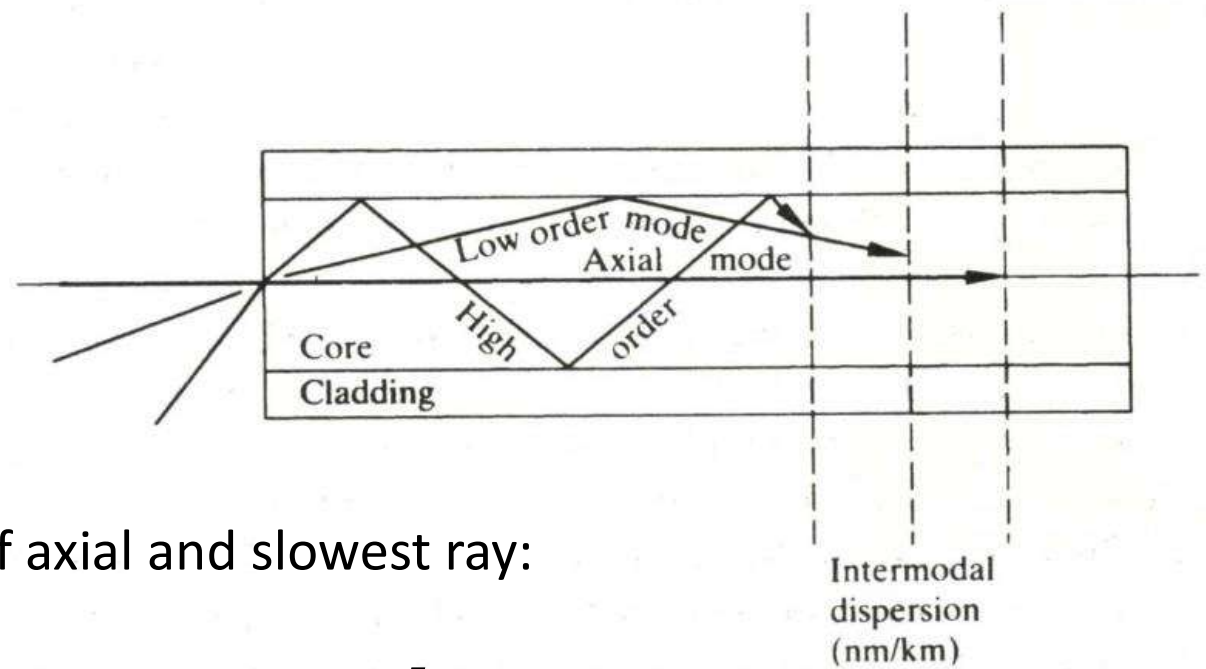
The stepped-index fiber was widely used in the first-generation systems (1975-1980)



Multimode fiber: depending on the launch angle into the fiber, there can be hundreds, even thousands, of different ray paths or modes by which energy can propagate down the core.



**Intermodal dispersion:** each mode in multimode fiber corresponds to a slightly different transit time.



Time delay between the arrive of axial and slowest ray:

Lowest order mode 
$$t_{min} = \frac{L}{v_f} = \frac{L}{c/n_f} = \frac{Ln_f}{c}$$

Highest order mode 
$$t_{max} = \frac{\ell}{v_f} = \frac{L/\cos \theta_t}{c/n_f} = \frac{Ln_f/n_c}{c/n_f} = \frac{Ln_f^2}{cn_c}$$

$$\Delta t = \frac{Ln_f}{c} \left( \frac{n_f}{n_c} - 1 \right)$$

## Example

The index of core and cladding  $n_f=1.5$  and  $n_c=1.489$

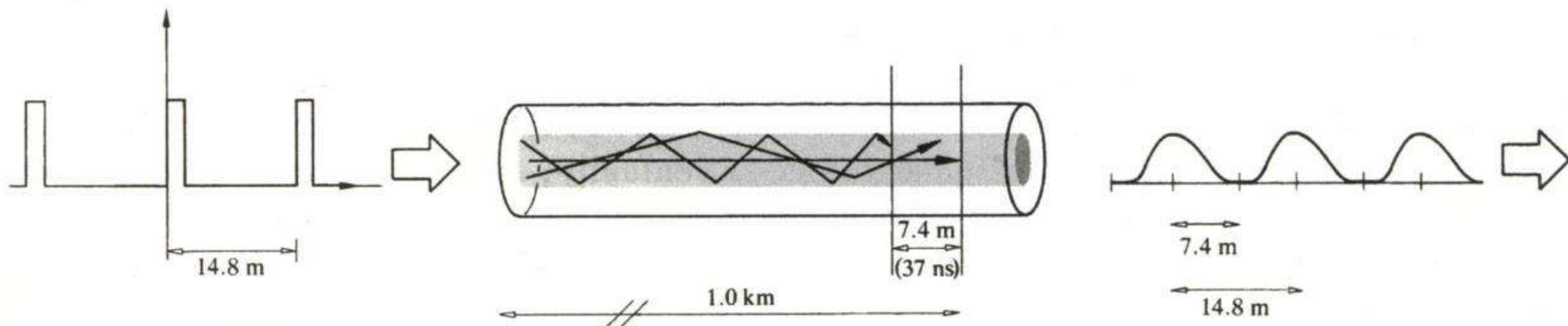
$$\Delta t = \frac{Ln_f}{c} \left( \frac{n_f}{n_c} - 1 \right)$$

The delay time between the lowest mode and highest mode:

$$\frac{\Delta t}{L} = 37 \text{ ns/km}$$

Traveling speed:  $v_f = \frac{c}{n_f} = 2.0 \times 10^8 \text{ m/s}$

The spread in space:  $7.4 \text{ m/km}$



It means that the input pulses must be at least 14.8 m apart; they must be separated in time by 74 ns and so cannot come any faster than one every 74 ns, which is a rate of 13.5 million pulses per second.

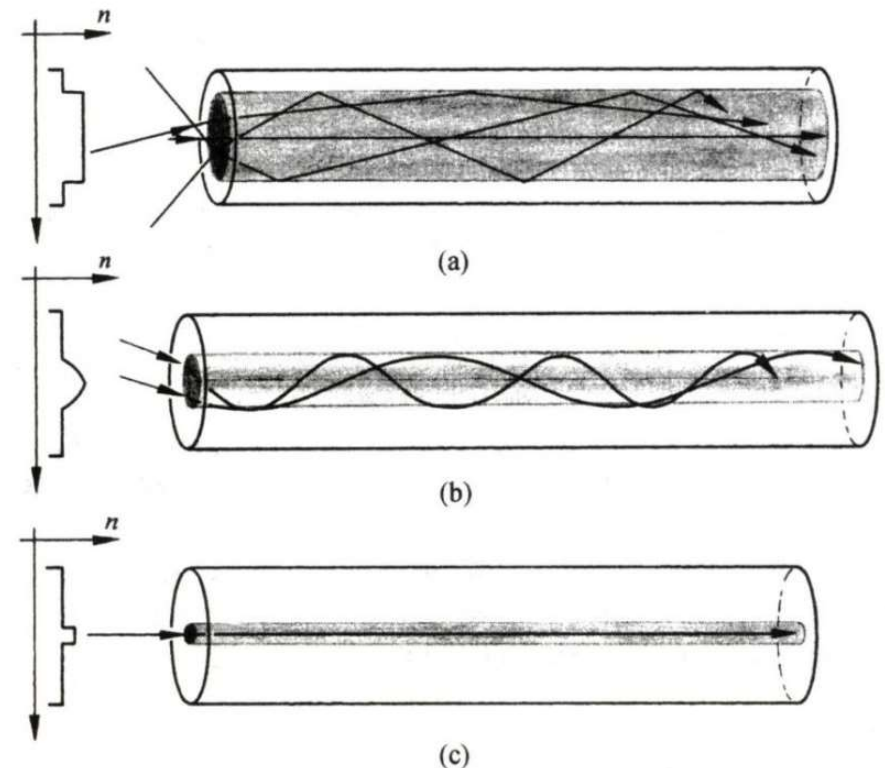
**Graded-index fibers:** refractive index decrease gradually from the core to the cladding.

Rays taking shorter paths are slowed down and rays spiralling around near the cladding move more swiftly over longer paths.

A graded-index fiber has a core diameter of about  $20\text{ }\mu\text{m}$  to  $90\text{ }\mu\text{m}$  and an intermodal dispersion of only around  $2\text{ ns/km}$ .

**Single-mode fibers:** the core is so narrow (less than  $10\text{ }\mu\text{m}$ ) that it will provide only one mode where the rays travel parallel to the central axis.

Typically having core diameters of only  $2\text{ }\mu\text{m}$  to  $9\text{ }\mu\text{m}$ , they essentially eliminate intermodal dispersion.



- (a) Multimode step-index fiber.
- (b) Multimode graded-index fiber.
- (c) Single-mode step-index fiber.

# Homework

□ 5.71, 5.73